



ADVIA Centaur[®] CP Immunoassay System

LIS Interface Guide

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Bayer HealthCare

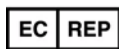
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1 Introduction

Who Should Read this Manual

The ADVIA Centaur[®] CP LIS Interface Guide is written primarily for software developers who are familiar with implementing layered communications protocols.

How this Manual is Organized

The following table provides an overview of the manual and explains the contents of each section:

<i>If you want to...</i>	<i>Refer to...</i>
review layered communications protocols and learn how they apply to the ADVIA Centaur CP system	Section 2, <i>Overview</i>
understand how the ASTM communications protocol applies to the ADVIA Centaur CP system, including messages and transactions for the data link and application layers	Section 3, <i>ASTM</i>
find out about the hardware interface specifications for the ADVIA Centaur CP system and the Laboratory Information System (LIS) device	Section 3, <i>ASTM</i>
learn about the communications package on the ADVIA Centaur CP system and how to interpret its screens	Section 3, <i>ASTM</i> Section 4, <i>Communication</i>
review examples of possible communications protocol errors and learn how to solve them	Section 5, <i>Troubleshooting</i>
review biohazard guidelines	Appendix A, <i>Safety Information</i>
contact Bayer HealthCare	Appendix B, <i>Support Information</i>
review examples of the communications protocol transactions and messages	Appendix C, <i>Messages</i>
review examples of the communications protocol records	Appendix D, <i>Records</i>
review invalid characters in ASTM messages	Appendix E, <i>Invalid Characters</i>
find a definition for an unfamiliar term	Appendix F, <i>Glossary</i>

Other ADVIA Centaur CP System Documents

For additional information on the ADVIA Centaur CP, refer to the following:

<i>This document...</i>	<i>Contains...</i>
<i>ADVIA Centaur CP Operator's Guide</i>	procedures and information about operating the ADVIA Centaur CP system.
ADVIA Centaur CP Online Help	detailed procedures and complete information about operating the ADVIA Centaur CP system; scheduled and as-needed maintenance procedures, troubleshooting procedures, and corrective actions for error conditions.
Calibrator and quality control product inserts	detailed information about each calibrator and control, including intended use, expected results, and performance data.

Conventions Used in this Manual

The ADVIA Centaur CP LIS Interface Guide uses the following text and symbol conventions:

Convention	Description
Bold	Bold type indicates the following: - text on the user interface. For example, if the word save appears as Save, it refers to the Save button on the user interface. - a key on the keyboard or a system key on the front of the ADVIA Centaur CP system. For example, when the word enter appears as Enter, it refers to the Enter key on the keyboard. - the actual text of a message transmitted between the ADVIA Centaur CP system and an LIS. - the actual value stored in the database.
<i>Italic</i>	Italic type refers to the title of a document or a section title in this manual. For example, <i>Overview</i> refers to Section 2 of this manual.
CAUTION	Caution statements alert you to conditions that may cause product damage or loss of data. On the system, this symbol indicates that you should refer to the manuals for more information.
NOTE	Note statements alert you to important information that requires your attention.

2 Overview

Layered Communications Protocols

The ADVIA Centaur CP system uses a three-layered communications protocol. Each layer performs a related subset of the functions the ADVIA Centaur CP system requires to communicate with another system.

- Layer 1 is the physical layer. This layer directs the transmission of the bit stream across the physical medium.
- Layer 2 is the data link layer. This layer transfers information across the physical medium. It sends blocks of data with additional synchronization and error control information to ensure a reliable transfer.
- Layer 3 is the application layer. This layer provides the information services of the ADVIA Centaur CP system.

Communication protocol standards are defined for each layer. For the ADVIA Centaur CP system to communicate with a laboratory information system (LIS), both systems must conform to the same three-layered protocol. The communications protocol used by the ADVIA Centaur CP system is an open, non proprietary communication standard developed by the American Society for Testing Materials (ASTM). If an LIS conforms to this protocol, it can communicate with any other system that also uses the standardized ASTM protocol.

<i>ADVIA Centaur CP System</i>	<i>LIS</i>
Layer 3: Application Layer Provides information services	Layer 3: Application Layer Provides information services
Layer 2: Data Link Layer Ensures that incoming and outgoing application messages are received and/or transmitted correctly	Layer 2: Data Link Layer Ensures that incoming and outgoing application messages are received and/or transmitted correctly
Layer 1: Physical Layer Transmits and/or receives the bit stream across the physical medium	

The physical layer and data link layer protocols are defined in *ASTM document E1381-95, Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems*. The application layer protocol is defined in *ASTM document E1394-91, Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems*.

Layers and Messages

The ADVIA Centaur CP system and the LIS are physically connected only at the physical layer and they logically communicate with each corresponding layer. The ADVIA Centaur CP application layer sends messages to the LIS application layer and receives messages from the LIS application layer. The same is true for the corresponding data link layers.

Messages are passed down or up from layer to layer, with each layer adding or removing its own protocol data to or from the given message. The term message is used generically to show that each layer has its own definition of a message. The following sections describe the application and data link messages.

Application Messages

At the application layer, the logical unit of transferred data is an application message. Application messages contain data that pertain to the computer system's application. An example of an application message sent from an LIS to the ADVIA Centaur CP system is an order to perform a specific test on a specific sample.

In the ASTM communication protocol, an application message consists of an ASTM header record, followed by any number of other ASTM records, such as patient and order records, and ends with an ASTM termination record. A simple ASTM application message that requests a T4 test for sample S19255 is shown below. Each ASTM record is a variable length record terminated by a carriage return character (<CR>).

```
H| \& |||Host||||ACCP1||P|1<CR>
P| 1 <CR>
O| 1 |S19255| |^^T4| R | ||||| O <CR>
L| 1<CR>
```

This application message consists of the following elements:

- H (header) record
- P (patient) record
- O (order) record
- L (termination) record

Refer to *ASTM document E1394-91* for a complete description of application messages. Refer to Section 3, *ASTM*, for a complete description of the implementation of the ASTM application layer by the ADVIA Centaur CP system.

When the ADVIA Centaur CP system receives the application message, as shown above, the system creates a worklist entry for the sample identified as A1234. This sample has a FSH test with routine priority. The worklist entry does not have any patient demographic information.



Data Link Messages

At the data link layer, the logical unit of data transferred or received is called a data link message. The data link message can consist of one character to thousands of characters. When the data link layer receives an application message to transmit as one message that contains multiple records, messages longer than 240 characters are divided into multiple intermediate frames followed by one final frame. Each frame is variable in length and has a maximum size of 240 characters that includes all data and protocol characters. The data link layer sends the message using as many frames as required.

Backus-Naur Form (BNF) notation is used to define a data link message that contains data link message symbols and ASCII mnemonics.

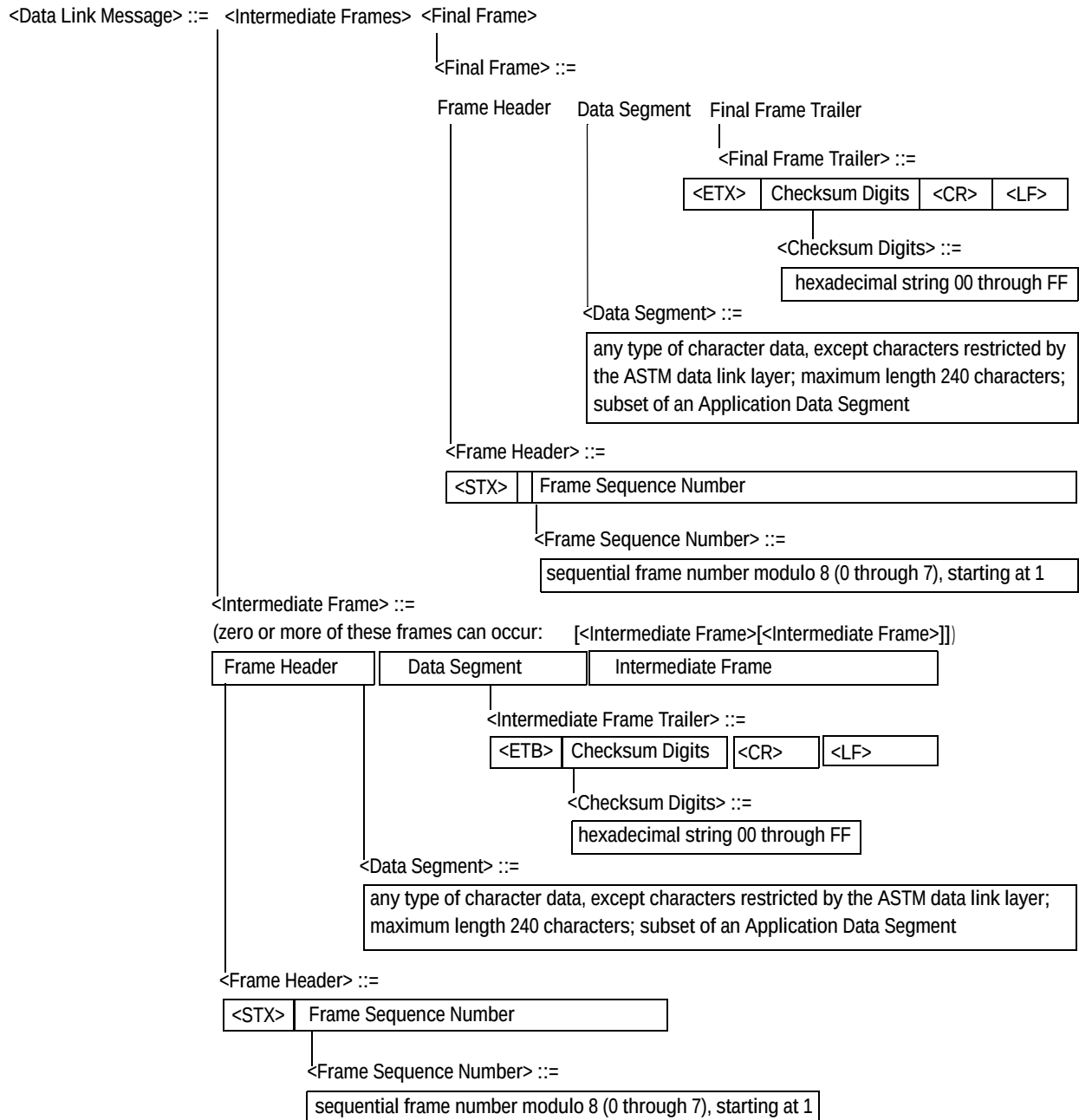
<Frame Header> ::= <STX> <Frame Sequence Number>

means

<Frame Header> is defined as an <STX> ASCII character followed by <Frame Sequence Number>

NOTE: The ASCII character mnemonics are not the same as data link message symbols. Data link message symbols contain multiple words of uppercase and lowercase letters and the ASCII mnemonics contain all uppercase letters.

The following figure shows the definitions of a data link message.



The ASTM data link protocol involves more than just adding and removing protocol characters to a given data stream. The ASTM data link protocol is a half duplex (ANSI definition) or simplex (CCITT definition) stop-and-wait protocol.

This means that either the ADVIA Centaur CP system or the LIS system can transmit data to each other, but only one at a time. Because of this requirement, the ASTM data link protocol has the following three handshaking phases that define an ASTM data link session:

- link establishment phase
- message transfer phase
- link release phase

The link establishment phase and the message transfer phase require a transfer of the character(s) initiated by the sender followed by a response by the receiver. The link release phase only requires a transfer by the sender.

The messages used in each phase of an ASTM data link session are shown below. A message transfer phase can consist of multiple data link messages. Refer to *ASTM document E1381-95* for a complete description of the ASTM data link layer specification.

Both the sender and the receiver send characters across the physical medium, but the sender is the only side that sends actual data. The data is contained in the data segments of the intermediate and final frames sent by the sender to the receiver.

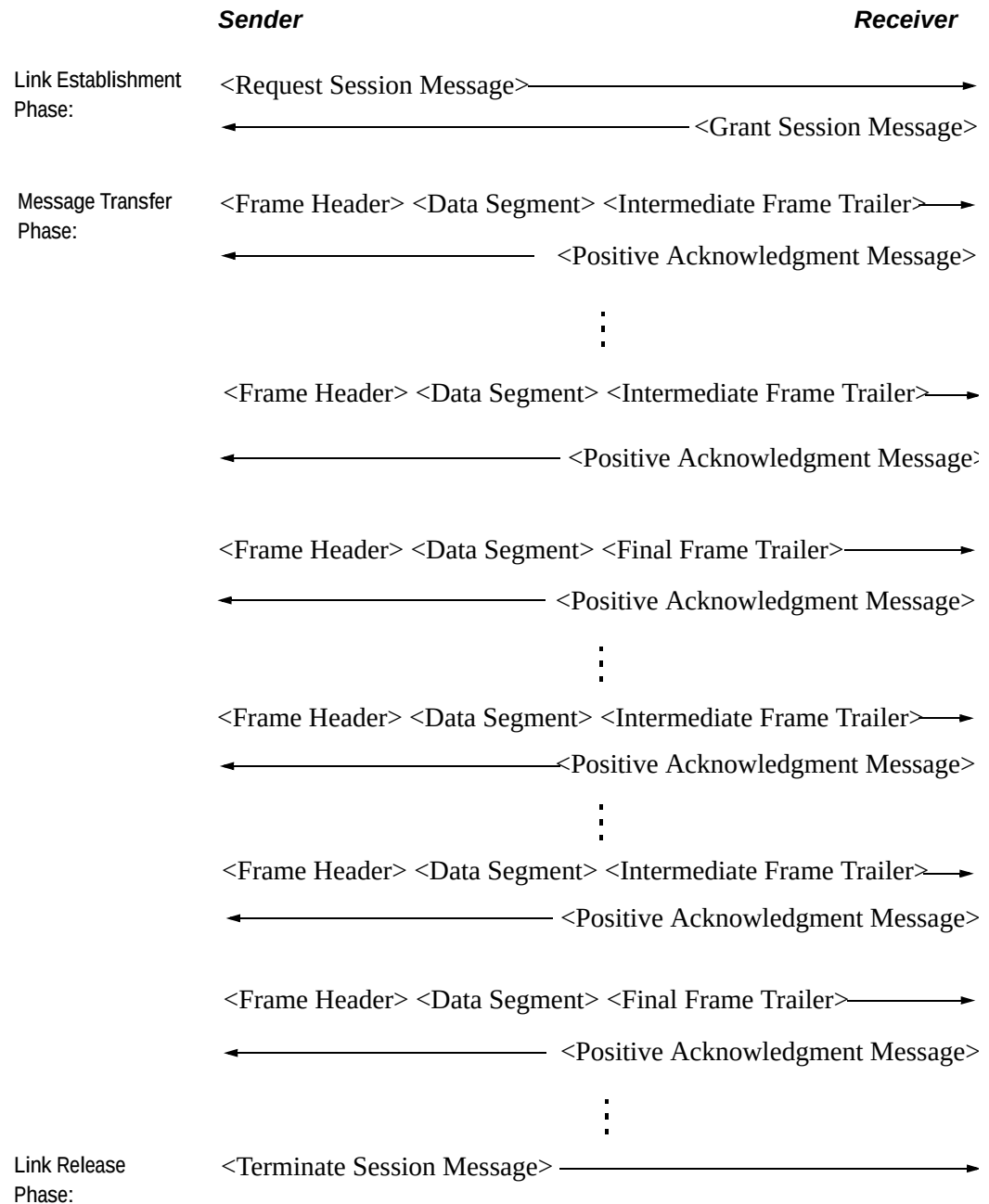
The sender and receiver terminology is based on the sender and the receiver of these data segments. Sender and receiver are abbreviated as S for sender and R for receiver.

Link Establishment Phase: S: <Request Session message> ::= <ENQ>
R: <Grant Session Message> ::= <ACK>
<Deny Session Message> ::= <NAK>

Message Transfer Phase: S: <Frame> ::= <Intermediate Frame> or <Final Frame>
R: <Positive Acknowledgment Message> ::= <ACK>
<Negative Acknowledgment Message> ::= <NAK>
or anything else except <ACK> or <EOT>
<Positive Acknowledgment with Interrupt Message> ::= <EOT>

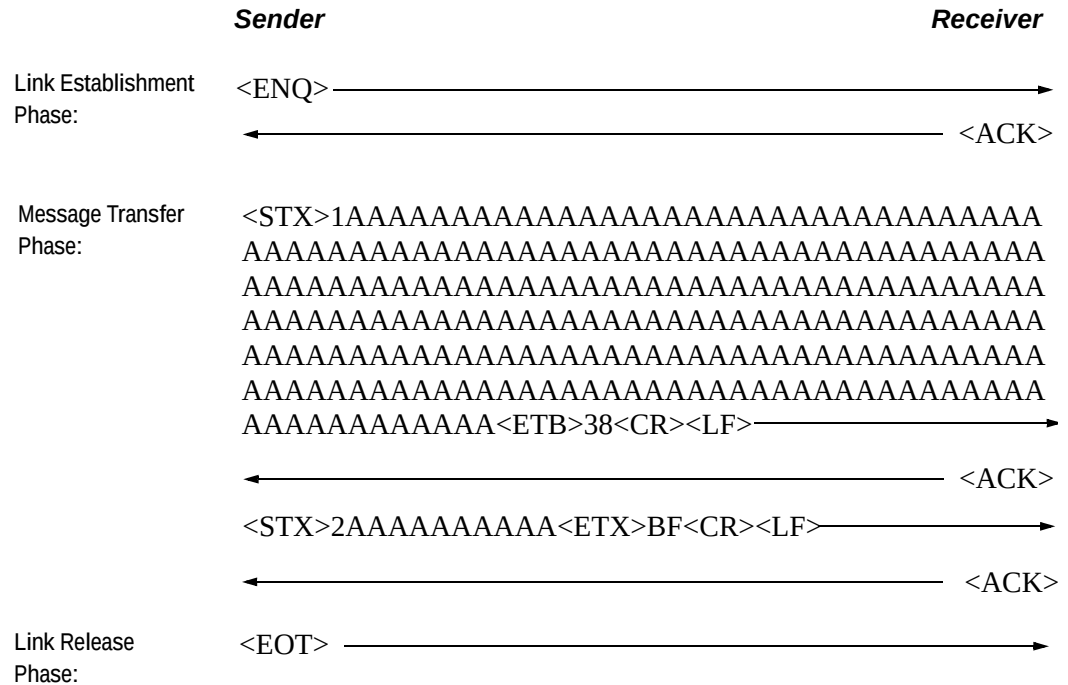
Link Release Phase: S: <Terminate Session Message> ::= <EOT>

The message definitions are used to establish an ASTM data link session. The following example shows an ASTM data link session using these messages.



For a complete description of the ASTM data link layer, refer to *ASTM document E1381-95*. This ASTM document explains ASTM data link layer protocol phases, message formats, timing constraints, restricted characters, error recovery, and a state transition diagram.

The following example shows an ASTM data link layer session with actual data. In this figure, an application message of 250 **A** characters needs to be sent from one system to another (250 is used since the maximum ASTM frame size is 240). Actual ASCII character mnemonics replace all of the ASTM data link symbols.



The following table describes the software events that take place at each phase:

Phase	Description
Link Establishment Phase	The data link layer of the sender begins the link establishment phase once it recognizes that it has an application data segment to transfer. The sender transmits a request session message (<ENQ>) and waits for a response. The receiver recognizes the received message as an <ENQ>. Because the receiver is idle, it sends a grant session message (<ACK>). The sender receives this <ACK>, which establishes the link between the sender and the receiver. The link establishment phase ends and the message transfer phase begins.
Message Transfer Phase	The sender determines that it must divide the application data segment into two frames. The first frame contains a frame header, a 240 character data segment, and an intermediate frame trailer. The sender transmits this frame and waits for a response. The receiver receives and validates the frame, transmits a positive acknowledge message (<ACK>), then waits for another frame. The sender receives the <ACK> and constructs the final frame. This frame contains a frame header, a 10-character data segment, and a final frame trailer. The sender transmits this frame and waits for a response. The receiver receives and validates the frame. The receiver transmits an <ACK>. The sender receives the <ACK> and enters the link release phase because the current application data segment was successfully transmitted and no other application data segments are queued.
Link Release Phase	The sender constructs a terminate session message (<EOT>) and transmits this message. The sender then returns to an idle state, waiting for an application data segment to transmit or receive. The receiver receives the <EOT> and also returns to an idle state, waiting for an application data segment to transmit or receive.

Physical Layer

The physical layer transmits and receives a bit stream of data across the physical medium. Transmitted data is passed down by the data link layer to the physical layer. The physical layer transforms this data into a sequence of electrical signals that are monitored by the physical layer at the remote system. The physical layer at the remote system transforms these electrical signals back into character form and sends these characters to the data link layer at the remote system.

The ASTM communications protocol uses a standard RS-232, point-to-point connection between the two systems. The ADVIA Centaur CP system supports various signal synchronization parameters, including the following:

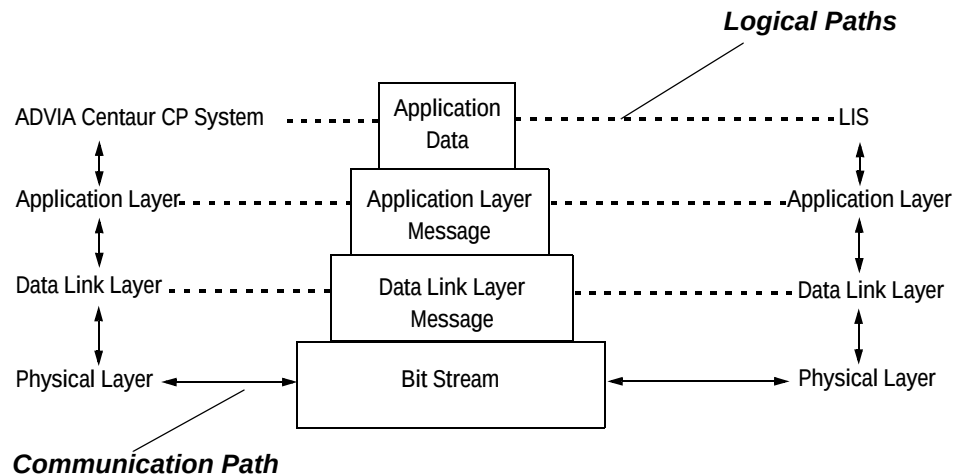
- five baud rates (19200, 9600, 4800, 2400, 1200)
- three parity bit checking schemes (Odd, Even, or None)
- 7 or 8 data bits
- 0 or 1 stop bits

Refer to *ASTM document E1381-95* for a complete description of the ASTM physical layer specification.

Interaction Between Layers

Each layer at one system logically communicates with the corresponding layer at the other system. For example, the application layer at the ADVIA Centaur CP system logically communicates with the application layer at the LIS by using application messages. However, the actual communication path is different than the logical communication path.

An application message is passed down to the data link layer and physical layer on the sending system, across the physical medium to the physical layer at the receiving system, up to the data link layer at the receiving system, and then up to the application layer at the receiving system. As the application message is passed through each layer, each layer adds or removes its corresponding protocol data.



You need to understand the functionality of the three layers of the ASTM protocol. You also need to implement a specific layer independently of the functionality of the other layers.

Each layer has its own purpose. The purpose of the application layer software is to generate and transmit messages to the other layers. The only purpose of the data link and physical layers is to transmit messages to the remote system.

In the example shown on page 7, the ASTM session transmitted 250 **A** characters. If this is what the application layer wants to transmit, the data link layer must attempt to transmit it. The data link layer needs to look at the data to determine the number of frames and to calculate checksums, but the software that performs data link layer services is not concerned with the construction or content of the application data segments.

Some implementation dependencies between layers do exist, such as the reporting of errors between adjacent layers, but this dependency is limited to the interface between the layers. In a special case required to configure some physical layer parameters, the application layer passes down the operator-selected physical layer parameters through the data link layer to the physical layer. Other than these few cases, the interaction between the application layer software, the data link layer software, and the physical layer firmware is highly cohesive and independent.

ASTM Application and Data Link Layer Interaction

The ASTM protocols defined for the application and data link layers do not specify what interface to use between these two layers. The ASTM application message and record format enables the application layer at the receiving system to handle application messages as a stream. Then the application layer at the receiving system can accept data from the data link layer at the receiving system on any data segment boundary. Also, the application layer at the transmitting system can transfer data to the data link layer at the transmitting system on any character boundary.

Interaction Between Layers on Incoming Messages

The application layer at the receiving system should process incoming data on a record by record basis. If the data link layer passes up only part of an application record, the application layer should not do anything with the partial record until the data link layer passes enough data segments to complete the record. Once the application layer receives a complete ASTM record, the application layer can process the record on the basis of the record type and the current status of the ASTM application message hierarchy.

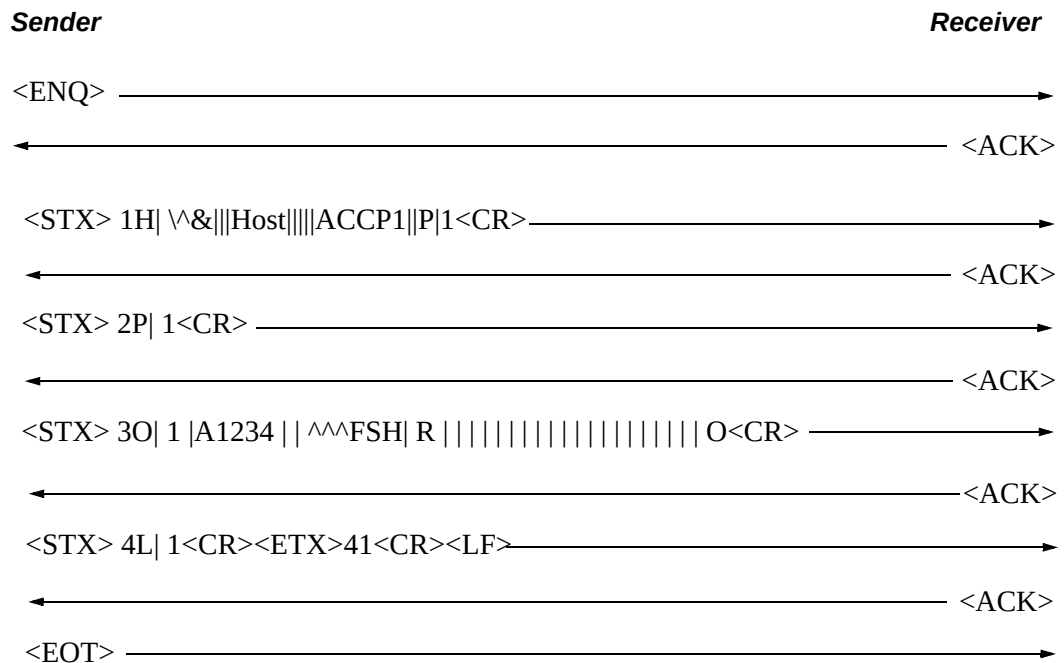
The ADVIA Centaur CP system ASTM data link layer passes data segments to the application layer as it successfully receives each intermediate or final frame. Because the ADVIA Centaur CP system ASTM application layer can accept application messages as a stream, the ADVIA Centaur CP system ASTM data link layer does not have to concatenate individual data segments from intermediate and final frames to rebuild the original application data segment. Handling concatenation at the data link layer requires an unknown amount of memory at the data link layer because a single application data segment can contain thousands of characters, but the longest data link layer data segment can have only 240 characters.

Implementations of Incoming Messages

The ADVIA Centaur CP system generally passes ASTM application messages to the data link layer on a message-by-message basis.

You can implement the generation of ASTM application messages on the LIS in a similar manner to the implementation of outgoing messages on the ADVIA Centaur CP system. Refer to *Implementation of Outgoing Messages*, page 2-12.

Your implementation is not required to conform to the message by message implementation on the ADVIA Centaur CP system. For example, another implementation is to pass the message down to the data link layer on a record-by-record basis. The following example shows an ASTM test order message session using this implementation. This method is suitable if there are memory constraints and an entire message cannot be completely generated before being transmitted.



Interaction Between Layers on Outgoing Messages

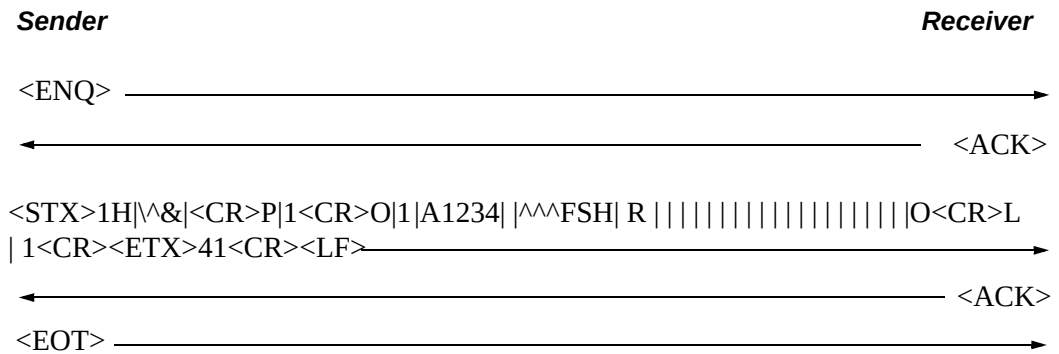
The software that processes incoming ASTM application layer messages must process these messages as a stream. The specifications, however, do not describe the interface between the ASTM application and data link layers on outgoing messages. Each implementation is responsible for defining its own interface.

Implementation of Outgoing Messages

Based on the type of message being generated, the outgoing application layer software constructs a single ASTM application message and passes it to the ASTM data link layer software. These messages generally consist of multiple ASTM application records. The ADVIA Centaur CP system transmits a separate message for the data in each sample.

An ADVIA Centaur CP system data link layer message can contain multiple ASTM application records within its intermediate frame and final frame data segment fields.

The following example shows a complete ASTM test order message session with a sequence of software events. The application message sent by the ADVIA Centaur CP system is the test order shown on page 2.



Event Log

Use the Event Log screen to identify errors and access online information about possible causes and corrective action procedures. Refer to the ADVIA Centaur CP Operator's Guide and Online Help for further information.

3 **ASTM**

Features

The ADVIA Centaur CP system can connect with other systems to exchange data. These other systems include a laboratory information system (LIS), and an ADVIA CentraLink[®] system.

The primary functions supported by the ADVIA Centaur CP system communications interface are receiving a worklist into the instrument and transmitting results from the instrument. These transfers can take place at the request of the ADVIA Centaur CP system operator or at the request of the remote system.

The ADVIA Centaur CP system can also send result data automatically. Automatic requests for worklist data are generated when the ADVIA Centaur CP system detects a new sample. The transfers are incremental (a single result or query for a single worklist entry) or batch. Batch and incremental transfers are supported between an ADVIA Centaur CP system and an LIS.

Operator-Initiated Transfers

The ADVIA Centaur CP system operator can initiate the transfer of result data from the ADVIA Centaur CP system to a remote system. The operator selects the records to transfer by using the results export function from the Result tab. The instrument sends the selected data to the other system, as long as communications are enabled.

Similarly, the operator can initiate the transfer of worklist entries from the remote system to the ADVIA Centaur CP system. The instrument sends a request for all worklist entries to the remote system, via the Download Order button in the Result tab, which responds by sending the requested data, as long as communications are enabled.

Remote System-Initiated Transfers

The remote system (an LIS) can send worklist entries to the ADVIA Centaur CP system. This transfer can take place at any time, if communications are enabled. Refer to *Data Link Layer Enabling and Disabling*, page 3-13, for more information.

The remote system can also send a request for result data to the ADVIA Centaur CP system. The request specifies the identifiers for the data requested (values of key fields). The ADVIA Centaur CP system responds by transferring the requested data. The remote system can request information at any time, if communications are enabled.

Automatic ADVIA Centaur CP System-Initiated Transfers

The ADVIA Centaur CP system can automatically transfer result data when the results are added to the result database. This transfer takes place if the automatic transfer option is enabled and the system is not configured to hold results at the Assay Setup screen. Each result is sent as it is generated. If a result cannot be sent when it is generated, the Centaur CP system tries to resend it together with the next result. If this fails too, it tries again with the next result, and so on.

The ADVIA Centaur CP system automatically transfers result data on hold when the operator releases the results.

The ADVIA Centaur CP system can automatically query for worklist entries from the LIS. This request occurs if the automatic worklist query option is enabled. Depending on the Query type selected, transfer occurs when the ADVIA Centaur CP system detects a sample being added to the sample compartment. The ADVIA Centaur CP system sends a request for patient demographics and tests to the LIS for that sample only.

Defining LIS Communication Options

Refer to the Screen Help section of the Online Help system for details on setting LIS Communications Options.

Requesting a Worklist from an LIS

The ADVIA Centaur CP system can obtain worklists from an LIS in the following circumstances:

- You select **System Automatically Queries Host for Worklist** at the LIS tab of the Setup screen and the user subsequently adds a sample to the sample compartment.
- You select **Download Orders** at the Result tab of the LIS screen.

Sending Results to an LIS

The ADVIA Centaur CP system can send results to an LIS under the following circumstances:

You select **System Automatically Sends All Results except Results on Hold** at the Setup – LIS tab and the system subsequently calculates a result.

There are three types of query modes to select from. See Appendix C for detailed examples:

- The Single query, the query is for a single request and may be followed by another single query, by an order record, or any other record.
- The second type of query is All, the query may be for multiple samples in one record and may be followed by any other record.

- The third type of query is ADVIA CentraLink. The query is for a single request and must be followed by an order record before another query can be made. If an order record is not received then the event is logged to the ACCP event log and after the query times out a new query record can be sent.

You select **Respond to LIS Query for Results** at the Setup – LIS tab and the system receives a query for results.

You can manually select and send

Limitations

The ADVIA Centaur CP system is an immunochemistry analyzer and not a data or communications manager. The communications functionality of the ADVIA Centaur CP system is secondary to its primary purpose, analyzing samples. The ADVIA Centaur CP system does not support certain communication features:

- managing a slave instrument
- routing messages between devices
- supporting more than one LIS at a time
- communications protocols other than the ASTM protocol

Physical Layer

This section describes electrical connections and signal characteristics of the physical interface. The interface uses a serial binary data interchange. The topology is point-to-point with a direct connection between the devices. The interface does not support multi-drop or network topologies. This physical interface conforms to ASTM specifications.

Electrical Characteristics

The electrical characteristics conform to the EIA-232-D-1986 (RS-232) Standard. A marking condition (binary one) corresponds to a voltage more negative than minus three volts with respect to signal ground. A spacing condition (binary zero) corresponds to a voltage more positive than plus three volts with respect to signal ground. The interface connection uses three contacts of a 9-position connector as shown in the following table.

Pin	Used for...
Pin 5	shield ground to ADVIA Centaur CP system frame; open at LIS.
Pin 3	transmitted data output from the ADVIA Centaur CP system.
Pin 2	received data input to the ADVIA Centaur CP system.

NOTE: An LIS connects to the ADVIA Centaur CP system through a serial RS-232 port. This RS-232 port has standard 9-pin, D-type, male connectors (DB-9P). Connect the ADVIA Centaur CP system to an LIS using a standard 9-pin, D-type, female connector (DB-9S).

NOTE: The RS-232 cabling should be a fully shielded type to minimize radio frequency emissions from the ADVIA Centaur CP system. When two systems are connected, connect one system on pin 1 and leave the other system open.

Signal Characteristics

The data transmission is asynchronous by character and serial by bit within a character. The order of bits in a character is as follows:

- one start bit (binary zero)
- character data bits (least significant bit first)
- parity bit
- stop bit(s) (binary one)

The circuit is in a marking condition between characters. The operator can select seven or eight character data bits and the parity bit as odd, even, or none. Odd parity corresponds to an odd number of one bits in the sequence of data and parity bits. Even parity corresponds to an even number of one bits in that sequence. The operator can also select one or two stop bits.

Transmission and reception operate at the same baud rate. The operator can select the baud rate. The interface allows selection of the following baud rates:

- 1200 baud
- 2400 baud
- 4800 baud
- 9600 baud
- 19200 baud

Data Link Layer

The data link layer provides an error-free exchange of data between two systems. It is not concerned with the contents of the exchanged data. The ADVIA Centaur CP system supports standard data link layer protocol as described in *ASTM document E1381-91*. This section describes the protocol supported by the ADVIA Centaur CP system.

ASTM Data Link Layer Protocol

The ASTM data link layer protocol provides methods for the following:

- link establishment
- message framing
- message frame sequence control
- link release
- flow control
- error detection and recovery

Link establishment and link release determine the sender and the receiver of data. Message framing ensures that the receiver recognizes the data. Message frame sequence control provides a mechanism to ensure that the system receives data in the correct order. Flow control allows the receiver to control the rate at which data is accepted. Error detection and recovery ensure that correct data is received.

The ASTM data link protocol is limited to half duplex transmission. This means that both sides can transmit, but only one at a time. The ASTM data link protocol transfers messages generated by the application layer. This protocol is not concerned with the content of the application messages. Some restrictions are placed on characters allowed in the data message. These restrictions avoid confusing application data with data link protocol control sequences.

Link Establishment

When data becomes available for transmission, the sender (either the LIS or the ADVIA Centaur CP system) attempts to establish control over the data link. If the link is already controlled, the message is queued for transmission. If the other system has control of the data link, the message must wait until the other system relinquishes control; otherwise, the system enters the establishment phase.

During the establishment phase, the sender requests control of the data link by transmitting a request session message. When the receiver detects a request session message, it determines whether it can receive data. The receiver can receive data if it has no data to send or if it is blocked from sending by a re-establishment delay. The sender must deal with four possible outcomes.

The first outcome is that the sender receives a grant session message from the receiver. The receiver has determined that it can receive data and grants control of the data link to the sender by replying with a grant session message. After transmitting the grant session message, the receiver enters the receive side of the message transfer phase. When the sender detects a grant session reply to the request session message, it enters the sender side of the message transfer phase.

The second outcome is that the sender receives a deny session message from the receiver. The receiver has determined that it cannot receive data. When the sender detects a deny session reply, it returns to the idle state and does not attempt to re-establish the data link for a specified period of time (governed by the busy timer). This delay allows the receiver time to clear the condition that caused it to deny the session. The receiver should deny a session when it cannot currently receive data. The busy timeout option is defined, in seconds, at the LIS tab of the Setup screen.

The third outcome is that the sender receives a request session message from the receiver. Contention has occurred; the receiver is attempting to start a session at the same time as the sender. The sender returns to the idle state and refrains from attempting to re-establish the data link for a specified period of time (governed by the contention delay timer). The other system detects the same situation and behaves similarly.

To avoid a second contention situation, the contention delay timers on the two systems are significantly different. The ADVIA Centaur CP system uses a one second delay, and assumes that the LIS uses a significantly longer delay. The current ASTM protocol describes suggested settings for this option for each system.

The fourth outcome is that the sender times out without detecting a grant session, deny session, or request session. Other characters or characters with errors such as parity errors are not recognized. The sender considers such replies as though no characters were received.

After a timeout occurs, the sender enters the termination phase (link release). The sender sends a terminate session message (EOT).

Message Transfer

The message transfer phase allows the sender to transmit data to the receiver. This transfer phase remains in effect as long as the sender has data to transmit and the receiver can accept more data. When the sender has transmitted all data, it enters the termination phase (link release).

Message Framing

Messages are sent in frames that consist of a frame header, a data segment, and a frame trailer. The frame header is a control sequence used to signal the start of the message and it contains a frame sequence number. The data segment contains the message text generated by the application layer. The frame trailer is a control sequence used to signal the end of the frame and it contains a frame checksum, a carriage return (<CR>), and a line feed (<LF>).

If the application frame text is longer than 240 characters, the text is split into multiple transmitted message frames. Text is copied from the application message buffer to the transmit buffer. If the entire application message text fits in a single frame, it is sent as a final frame message. If the entire application message text does not fit in a single frame, the part of the message that does fit is sent as an intermediate frame message.

When the intermediate frame message is successfully transferred to the receiving side, more of the message text is copied to the transmit buffer. This process repeats until all of the application message text is received, the last frame being a final frame.

The data link layer receives the outgoing application message as one message. Then the data link layer sends the message using as many frames as required.

Frame Sequence Numbers

Frame sequence numbers are used to assure that data is accepted in the correct order by the receiver. A sequence number is formatted as the ASCII representation of its decimal value in one digit.

The first message after link establishment is given the initial sequence number of one. All subsequent frames (intermediate or final) are given sequence numbers computed by adding one to the previous sequence number. If the number exceeds the high value of seven, it resets to the low value of zero (modulo 8 arithmetic). A retransmitted frame is sent with its original sequence number. After link release and a new link establishment, sequence numbers start again with the initial sequence number.

Frame Checksum

Frame checksum is used to detect errors in the frame that are not detected by character parity checking. Its coverage ranges from the second character of the header (the frame sequence number) to the last character of the trailer before the checksum itself. The algorithm used is binary sum, formatted as the ASCII representation of its hexadecimal value (modulo 256) in two digits with leading zeros, a range of 00–FF.

Frame Acknowledgment

When a receiver receives a complete frame, it responds to the sender with an acknowledgment. If the frame contains an error, the receiver sends a negative acknowledgment message. Refer to *Error Detection and Recovery*, page 3-8, for more information on frame errors. If the frame is valid, the receiver sends one of two positive acknowledgment messages:

- a positive acknowledgment with interrupt message, if the receiver has data to send
- a positive acknowledgment message

If the transmitting system receives a positive acknowledgment with interrupt, the system can enter the termination phase (link release). If the transmitting system releases the link, the system should wait for at least 15 seconds or wait for the other system to initiate and terminate a session.

If the transmitting system does not receive a positive acknowledgment with interrupt for the last frame or if the system does not wish to release the link, the next frame is started. When there are no additional frames to transmit, the transmitting system enters the termination phase (link release).

When the sender detects a negative acknowledgment, it increments the retry count. If it sent the maximum number of retries (six) for one frame, the message is noted as not sent, and the sender enters the termination phase (link release). If the sender does not exceed the maximum number of retries, the sender retransmits the frame. If the sender does not detect a reply within the no response time interval, the sender considers the remote system down, and then enters the termination phase (link release). If the sender receives a character other than a positive acknowledgment, positive acknowledgment with interrupt, or negative acknowledgment, the sender assumes a negative acknowledgment.

Link Release

The link release phase returns the data link to the idle state. When the sender transmits a terminate session message, it enters its idle state. When the receiver detects a terminate session message, it enters its idle state. If the receiver has data to transmit, it can start link establishment.

Error Detection and Recovery

Several types of communications errors are detected with the ASTM protocol:

- character errors
- checksum errors
- sequence errors
- timeouts

Character errors are detected on individual characters in a message. They include parity, framing, and overrun errors. These errors result in the invalidation of the individual character and the data link frame that contains the character. Any frame containing a character error is considered a bad frame and is responded to with a negative acknowledgment.

Checksum errors occur on messages that contain checksums. If the computed checksum does not match the received checksum, one or more of the characters in the message had an undetected parity error (two-bit error). These errors result in the invalidation of the frame. Any frame containing a checksum error is considered an invalid frame and is responded to with a negative acknowledgment.

Sequence errors occur when the sequence numbers of frame messages are not the next number in sequence. If the sequence number is the number of the previously accepted and acknowledged frame, the frame is assumed to be a retransmission and positively acknowledged. Otherwise, the sender and receiver are out of synchronization. Any frame containing a frame synchronization error is considered an invalid frame and is responded to with a negative acknowledgment.

Timeouts can occur whenever one system is waiting for the other to do something specific. These conditions include:

- receiver waiting for a frame
- sender waiting for a reply to a session request
- sender waiting for a reply to a frame

If the timeout occurs when the receiver is waiting for a frame (governed by the interframe timer), it assumes the sender is no longer operating. The receiver returns to its idle state.

If the timeout occurs when the sender is waiting for a reply to a session request (governed by the no response timer), it enters the termination phase.

If the timeout occurs when the sender is waiting for a reply to a frame (governed by the no response timer), it assumes the receiver is no longer operating. The sender enters the termination phase (link release).

When the data link layer fails to successfully send a message (bid failure, excessive retries), it notifies the application layer. A status code indicates the reason for the failure. The application layer handles and logs the error.

Flow Control

The ASTM data link protocol does not support XON/XOFF flow control. The ADVIA Centaur CP system uses an acknowledgment delay to allow itself more time to process data already received.

Flow control is used by the receiver to limit the rate of sending characters. When the receiver detects that its receive buffers are filling up after receiving a frame, the receiver can wait for a specified time before sending the positive or negative acknowledgment message.

ASTM Data Link Layer Protocol Events

The ASTM data link layer protocol consists of several types of events:

- messages
- timers
- frame limits
- miscellaneous characters

The ASTM data link protocol events are defined in the following tables.

ASTM Data Link Messages - Sender

Event	Definition
request session message	The potential sender uses the request session message to gain control of the data link. ASTM protocol defines it as the single character <ENQ>, ASCII value of 5.
intermediate frame message	All but the last packet of text from a data link message uses the intermediate frame message. This message contains a sequence number and a checksum. It must contain message text. ASTM protocol defines it as beginning with the character <STX>, ASCII value of 2, followed by a one-character frame sequence number (refer to <i>Frame Sequence Numbers</i> , page 3-7). The data segment is next, with 1–240 ASCII characters, no restricted characters allowed. After the data segment is the character <ETB>, ASCII value of 23, which is followed by a two-character frame checksum (refer to <i>Frame Checksum</i> , page 3-7). Next is the carriage return character <CR>, ASCII value of 13. The frame ends with the line feed character <LF>, ASCII value of 10.
final frame message	The final frame message is used for the last packet of text from a data link message. It contains a sequence number and a checksum. It must contain message text. ASTM protocol defines it as beginning with the character <STX>, ASCII value of 2, followed by a one-character frame sequence number (refer to <i>Frame Sequence Numbers</i> , page 3-7). The data segment is next, with 1–240 ASCII characters, no restricted characters allowed. After the data segment is the character <ETX>, ASCII value of 3, which is the only difference between a final frame and an intermediate frame. Next is a two-character frame checksum (refer to <i>Frame Checksum</i> , page 3-7). Next is the carriage return character <CR>, ASCII value of 13. The frame ends with the line feed character <LF>, ASCII value of 10.
terminate session message	The terminate session message is used by a sender to relinquish control over the data link. ASTM protocol defines it as the single character <EOT>, ASCII value of 4.

ASTM Data Link Messages - Receiver

<i>Event</i>	<i>Definition</i>
grant session message	The receiver uses the grant session message to grant control of the data link to the requester. ASTM protocol defines this message as the single character <ACK>, ASCII value of 6.
deny session message	The receiver uses the deny session message to deny the request to take control of the data link. ASTM protocol defines this message as the single character <NAK>, ASCII value of 21.
positive acknowledgment message	The receiver uses the positive acknowledgment message to notify the sender that the last frame was successfully received. ASTM protocol defines this message as the single character <ACK>, ASCII value of 6.
positive acknowledgment with interrupt message	The receiver uses the positive acknowledgment with interrupt message to notify the sender that the last frame was successfully received and the receiver has data to send. ASTM protocol defines it as the single character <EOT>, ASCII value of 4.
negative acknowledgment message	The receiver uses the negative acknowledgment message to notify the sender that the last frame from the sender was not successfully received. Using the ASTM protocol, the sender uses any message other than a positive acknowledgment (with or without interrupt) as a negative acknowledgment message. ASTM protocol defines it as the single character <NAK>, ASCII value of 21.

Senders and receivers use timers to provide blocking delays following unsuccessful line bids and to recover from a failure (watchdogs) on the part of the other system. Each timer uses one value in seconds.

ASTM Data Link Timers - Receiver

<i>Timer</i>	<i>Definition</i>
interframe timer	The receiver uses the interframe timer to limit the amount of time it waits for the next frame (or session termination) following session establishment or after sending an acknowledgment to a previous frame. If this timer expires, the receiver returns to an idle state.

ASTM Data Link Frame Limits

<i>Event</i>	<i>Definition</i>
frame retry limit	The frame retry limit is the maximum number of attempts the sender can retransmit a frame before it terminates the session. ASTM protocol uses 5 retries for a maximum of 6 attempts.
frame text size limit	The frame text size limit is the maximum number of message text characters allowed in a data segment of a single frame. ASTM protocol uses 240 characters as the limit.

ASTM Data Link Miscellaneous Characters

<i>Character</i>	<i>Definition</i>
restricted characters	Restricted characters are those characters not allowed in the data segment text. If any restricted characters are present, the message is not sent. The ASTM data link protocol restricts the characters with ASCII values of 1, 2, 3, 4, 5, 6, 10, 16, 17, 18, 19, 20, 21, 22, and 23. Although the protocol permits the null character (ASCII value 0), the ADVIA Centaur CP system considers this character a restricted character due to its common usage as a string termination character. The 8-bit characters with ASCII values of 129–256, are mapped to their 7-bit equivalents.

Data Link Layer Enabling and Disabling

The data link layer is enabled or disabled when you turn on or off the ADVIA Centaur CP system. When an operator selects the On radio button in the Connection frame or disables communication by selecting the Off radio button at the LIS tab of the setup screen.

The screenshot shows the 'LIS' tab of the ADVIA Centaur CP LIS Interface Setup screen. The interface is divided into several sections:

- Identification:** Includes fields for 'Instrument' (set to 'ACCP1') and 'Host'.
- Communication:** Includes dropdown menus for 'Comm Port' (3), 'Baud Rate' (9600), 'Data Bits' (8), 'Stop Bits' (1), and 'Parity' (None).
- Delimiters:** Includes text boxes for 'Field' (|), 'Repeat' (\), 'Component' (^), and 'Escape' (&).
- Timeout:** Includes numeric input fields for 'Query' (60), 'No Response' (15), 'Interframe' (30), and 'Busy' (10), each followed by the unit 'seconds'.
- Connection:** Includes radio buttons for 'On' and 'Off'.
- Log File:** Includes checkboxes for 'Record ASTM 1381' and 'Record ASTM 1394', each with 'On' and 'Off' radio buttons.
- Transmission Options:** Includes a checkbox for 'System Automatically Queries Host for Worklist', a 'Query Mode' dropdown (set to 'Single'), a checkbox for 'System Automatically Sends All Results except Results on Hold', a checkbox for 'Respond to LIS Query for Results', and a checkbox for 'Remove Leading spaces from SIDs'.

At the bottom right, there are 'Save' and 'Cancel' buttons.

Application Layer

This section describes the protocol that the ADVIA Centaur CP system uses to transfer data between two systems and to process that data. This section enumerates the actual transfers allowed and the types of processing that take place on the data.

You can transfer worklist information to the ADVIA Centaur CP system to an LIS. You can also transfer results from the ADVIA Centaur CP system to the LIS.

The application layer provides services to a remote system so that it can access this data.

Data Transfer and Processing

Data is transferred from a source system to a destination system. If the ADVIA Centaur CP system is the source, then the LIS is the destination, or if the LIS is the source, then the ADVIA Centaur CP system is the destination.

ADVIA Centaur CP System as Data Source

When the ADVIA Centaur CP system is the source of worklist entries and result data, it can perform the following actions:

- respond to remote requests for a worklist or result data
- initiate the transfer of results automatically or by operator command
Automatic transfer occurs when new results are available (instrument readings or operator-entered), the operator has enabled the automatic results transfer option, and the results are not on hold.
- initiate requests for worklist data automatically or by operator command

Selecting Data for Transfer

When data (worklist entries or results) is transferred to an external system, a subset of the entries is selected from all of the available data. This section describes the selection process, the database keys used to select the records, and the data required to perform the selection.

The selection criteria are restrictive. The absence of criteria is treated as requesting all records. Invalid selection criteria are treated as no restriction.

The value of the selected fields is specified as a single value, a list of values, or a range, although all fields are not specified each way. If a single value or a list is given, the system selects only those entries that exactly match one of the values. If a list is given for a field that has multiple components, the system considers each component as an independent list. If the list uses more than one component in the list, an entry must have a match in each transferred list. If a range is given, the system selects those replicates whose field value falls within the range. Ranges in request for data messages (queries) are either open-ended or closed. If only a starting value is given, the system sends all entries with a field value greater than or equal to the specified value. If only an ending value is given, the system sends all entries with a field value less than or equal to the specified value. If a starting and an ending value are given, the system sends all entries within the range. All alphabetic character comparisons are case-sensitive.

If **System Automatically Sends All Results except Results on Hold** is selected at the LIS tab of the Setup screen, the system sends results. The system does not send calibrator results and invalid results from samples. Examples of results include:

- unmeasured replicate results
- results of all of the levels in a dilution profile
- dilution results that are out of range or overdiluted
- results that the system cannot calculate because the RLUs are above or below the Master Curve
- results of the tests used to calculate the result of a ratio test.

Selecting Result Entries

Results entries are selected from the result database. Several fields are used to restrict the choice of entries. The selected entries are based on the combination (logical AND) of the selection criteria. The default selection is all entries.

The system does not send all result entries. It does not send calibrator results, only patient sample and control results.

A request for a data message (query) from a remote system can use the Specimen ID field.

The Specimen ID describes one or more ranges. If you specify multiple ranges, the system selects entries based on the logical OR of the ranges. Refer to *Query Record*, page 3-41, for more information.

Selecting Results by the Operator

The operator can only use the export button in the Result tab to select results to be manually transmitted to the LIS.

Selecting Results by the Remote System

A request for data message query from a remote system can use the Specimen ID selection criteria.

The Specimen ID describes one or more ranges. If multiple ranges are specified, the system selects replicates based on the logical OR of the ranges. Refer to *Query Record*, page 3-41, (check) for more information.

ADVIA Centaur CP System as Data Destination

When the ADVIA Centaur CP system is the destination of worklist entries, it can perform the following actions:

- request a worklist from the remote system by operator command
- accept a remote worklist transmission
- automatically request worklist data from the remote system during sample processing
The system transmits an automatic request when it detects a new sample being loaded in the sample compartment.

As samples are loaded into the sample compartment, the system requests information for the newly added samples by using one query for each sample. Each query receives a reply or is canceled before the next query is issued. The operator enables or disables this option via System Automatically Queries Host for Worklist.

When using the Download button on the Results screen, the worklist request uses a single query to ask for all available worklist entries.

The system accepts a worklist from a remote system without having requested it. The system accepts unsolicited messages when communications are enabled. Refer to *Data Link Layer Enabling and Disabling*, page 3-13, for more information.

Processing Received Data

When the system receives data from an external system, it internally processes the data before adding it to the system database. This section describes this process and the data required to perform it.

Processing Worklists

When the system receives a worklist entry from an external system, it checks the entry against the worklist database. Matching is based on SID. If the SID field is not present in the external entry, the system rejects the entry. If the SID field is present, the system uses it to search the database. If the worklist entry is not found, the system accepts the data and creates a new record in the database.

If the worklist entry is found, the system uses the external data to update the database record. The updating is also subject to order action codes. Refer to *Order Record*, page 3-28 for more information.

If the order is canceled, the worklist entry is deleted. If the order is not cancelled, the patient record in the database is updated. If the testorder already exists and the result is pending, the new workorder is rejected.

When the system uses external data to update an existing record in the database, it uses any non-null incoming fields pertaining to patient demographics or sample identification to overwrite existing data. Incoming fields with the ASTM erase sequence ("") erase the field in the database. The system ignores null incoming fields.

The action code of the order record determines whether to select or deselect the test in the worklist database. If the test is complete, the system cannot deselect it, and takes no action. If the system recognizes any of the incoming tests, the system adds patient demographic data and recognized tests to the worklist. If the system does not recognize some of the tests, it returns an event message, and adds the recognized tests to the worklist. If the system recognizes none of the tests, it does not change the worklist. This allows the remote system to send a blanket worklist to several instruments with each running only those tests it recognizes.

If the matching causes an update of the database (change to existing record) and the system is currently processing the updated sample, the test scheduler is notified.

Processing Results

The ADVIA Centaur CP system ignores any results received from an external system.

Application Layer Protocol

The ADVIA Centaur CP system application layer protocol conforms to the E1394-91 ASTM specification, *Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems*. This section describes how messages are structured and identifies the contents of the message fields. This section also explains minor exceptions to the protocol.

Messages

Messages consist of a series of hierarchically structured records. The records are:

- header record (level 0)
- patient record (level 1)
- order record (level 2)
- result record (level 3)
- query record (level 1)
- comment records (levels 1 through 4)
- termination record (level 0)

A message begins with a header record and ends with a termination record. The protocol allows multiple patient, and query records within a message. Multiple order records may occur with each patient record. Multiple result records may occur with each order record. Multiple comment records may occur at any place in the message. They are associated with the immediately preceding record other than another comment record.

The ADVIA Centaur CP system attempts to conform to the E1394-91 ASTM standard in its most general form. On incoming messages, the system allows the intermixing of worklist entries, results, and queries. Normally, only one type of transaction occurs in a given message. In general, level 1 records (patient and query) correspond to the specific transactions (database update or query). Patient records are updates to the worklist. The action that takes place depends on the existence of subsidiary records (order, result) and the status or action codes in them. Query records initiate a response. The response is normally to send worklist entries or result data.

The ADVIA Centaur CP system scans all record types on incoming messages. The system scans and validates all record sequence numbers. Any invalid record types or any invalid sequence numbers cause an error. All fatal parse errors immediately close the current application message. The system processes any records received prior to the record containing the fatal parse error. The system ignores scientific records on all incoming messages. In order records, if the error is localized, the system rejects only the affected test order. If the error extends to the entire order record, the system rejects the whole record. The system rejects all result records. The system ignores manufacturer's records.

Except for the header record, each record in an ASTM message has a sequence number that reflects the hierarchical structure of the message. The system uses the number that is the *n*th occurrence of the record type at the same hierarchical level. This number is reset to 1 whenever the system transmits a record of greater hierarchical significance (lower number) or if the system uses the same record at a different hierarchical level, such as comment and manufacturer's records. Comment records have no explicit level, but the system considers them as one level of significance less (higher number) than the associated non-comment record.

The following table shows an example of a message structure.

Record Type	Hierarchical Level	Sequence Number
Header	0	no sequence number
Patient	1	1
Comment	2	1
Order	2	1
Result	3	1
Comment	4	1
Result	3	2
Result	3	3
Comment	4	1
Comment	4	2
Patient	1	2
Order	2	1
Result	3	1
Query	1	1
Patient	1	3
Order	2	1
Result	3	1
Termination	0	1

The following table describes the types of messages the ADVIA Centaur CP system sends and receives.

Message Type	Description
worklist message	The worklist message consists of a header record, a patient record for each worklist entry, a number of optional patient comment records for each entry, one or more order records for each entry, and a termination record. A worklist message is a response to a query.

Message Type	Description
result data message	The result data message consists of the following records: • a header record • a patient record for each sample • a number of optional patient comment records for each sample • one or more order records for each sample • one or more result records for each test ordered • a number of optional result comment records for each result record • a termination record A result data message is a response to a query, automatically initiated by the system, or is operator-initiated.
response to query in error	A response to a query that contains an error consists of a header record and a termination record. The other system should respond by closing the query and logging the error or notifying the operator of a query in error.
response to messages in error	A response to a message that contains an error, such as ordering an unknown test or a patient record with no specimen ID, consists of all records in the hierarchy up to the record in error, followed by a comment record, and a termination record. The comment record describes the error.
cancellation of query message	A cancellation of query message consists of a header record, a query record, and a termination record. The other system should respond by canceling any transmission in progress that is in response to the canceled request. Then the other system sends a message that ends with a termination record. The termination record has a status indicating the message was canceled at the receiver's request.

ASTM Record Structure

ASTM protocol uses 8-bit characters with values 7, 9, 11, 12, and 13 and values in the range of 32–126, which are defined by the ASCII standard (ANSI X3.4-1986). The ADVIA Centaur CP system permits the transfer of values in the range of 129–254, but transfers them to their corresponding 7-bit values. The ASCII value 13 (carriage return) is reserved as a record terminator. The record structure does not allow unused values in ASTM records, but transmits the unused values through the use of escape sequences supported by ASTM.

ASTM records are composed of fields separated by field delimiters; the character **|** is the default field delimiter. Each field is identified by its position in the record, which is determined by the number of field delimiters that precede a field. Fields are repeated if they are separated by repeat delimiters; the character **** is the default repeat delimiter. If there are no characters between two field delimiters, the field is empty (null). A null field is an indication that a corresponding field in the database of the system receiving the record should remain unchanged or that the field is unused. A field that contains two double quotes ("") indicates that the system receiving the record should delete the corresponding field in the database (database field is emptied). Any fields after the last field containing data can be left out of the record, that is, the system truncates unused fields from the record. The system assumes any field not present in the record to be null. An ASCII carriage return is used to indicate the end of the record.

Fields are separated into components, separated by component delimiters; the character **^** is the default component delimiter. Individual field components are not repeated. Only the entire field is repeated. Component delimiters are only required up to the last non-empty component. The system truncates unused components from the field. Empty or absent components are null and indicate that a corresponding component or field in the database of the receiving system should not change or that the component is unused. A pair of double quotes in a component does not delete the component. Double quote pairs mean deletion only at the field level.

Escape sequences are supported by ASTM and use the character **&** as the default escape delimiter. The ADVIA Centaur CP system does not support the escape sequences to start highlighting text (**&H&**), stop highlighting text (**&N&**), or the manufacturer defined escape sequence (**&Zcccc&**). The ADVIA Centaur CP system does not send such escape sequences and, if they are received, removes them from data.

The ADVIA Centaur CP system supports the ASTM escape sequences for delimiters and hexadecimal data. The delimiter escape sequences allow embedding the delimiter characters in data without interpreting the characters as delimiters. The characters **&F&** represent the field delimiter, the characters **&S&** represent the component delimiter, the characters **&R&** represent the repeat delimiter, and the characters **&E&** represent the escape delimiter. The characters **&Xhhhh&** represent the hexadecimal data escape sequence, begun by **&X**, followed by any number of pairs of hexadecimal digits (0–9, A–F), and ended by the escape delimiter (**&**). If there is an odd number of hexadecimal digits, the system considers the last hexadecimal digit as the least significant digit of a pair of hexadecimal digits. For example, the characters **&XA&** represent the ASCII linefeed character. The system converts each pair of hexadecimal digits to an ASCII value. For example, the system converts characters **&X40&** to the ASCII character **@**.

If an escape sequence converts to an ASCII character not allowed in the ADVIA Centaur CP database, the system replaces the character with a single ASCII space character. The ASCII characters allowed are 32–126, 161–163, 165, 167–171, 176–179, 181–183, 185–189, 191–207, 209–221, 223–239, and 241–253. If the database field of the ADVIA Centaur CP system is more restrictive (date and sex fields), the system considers the field in the record as null.

The contents and field sizes of records generated by the ADVIA Centaur CP system and the worklist and results databases define the records accepted by the ADVIA Centaur CP system. Refer to page 21 through page 48 for maximum sizes. The record descriptions indicate the source and special formatting of the field contents. The field number within the record is listed as T#, where T is the record type and # is the field number. Field, repeat, component, and escape delimiters are the characters |, \, ^, and &, respectively. Fields not listed are null.

Record Data Contents

This section describes the contents of record data fields for both incoming and outgoing (generated by the ADVIA Centaur CP system) records. It provides an example of each record. For more record examples, refer to Appendix D, *Records*.

Header Record

The header record is always the first record in an application message. It contains general sender and receiver information. Its hierarchical level is 0. The following table describes the values each field can contain.

Field	Field Name	Value
H1	Record Type	Outgoing: 1 character; always H . Incoming: 1 character; can be H or h .
H2	Delimiter Characters (Field, Repeat, Component, Escape)	Outgoing: Always ^\&, respectively. Incoming: 0–4 characters; default ^\&. Recommend using ^\& to match ADVIA Centaur CP system outgoing delimiters, but can use any 4 of the ASCII characters allowed.
H5	Sender ID	Outgoing: 1–10 characters; default ACCP1 . Set to System Name at the Setup – LIS Tab. Incoming: 0–10 characters. Not used for incoming message processing.

Field	Field Name	Value
H10	Receiver ID	Outgoing: 1–10 characters; default Host Set to LIS ID field at the Setup – LIS tab. Incoming: 0–10 characters. The Receiver ID is used to ensure that the received message was addressed to the ADVIA Centaur CP system. If the Receiver ID field is empty, the ADVIA Centaur CP system assumes that it is to process the received message. If the Receiver ID field exists and is not null, the ADVIA Centaur CP system compares it to the System Name at the Setup – LIS tab. If the two fields are equal (case-sensitive comparison, for example, A does not equal a), the ADVIA Centaur CP system accepts the message for processing. If the two fields are different, the ADVIA Centaur CP system rejects the entire application level message, and returns an error message to the LIS.
H12	Processing ID	Outgoing: Always 1 character; P (Production) Incoming: Not processed
H13	Version Number	Outgoing: Always 1. Incoming: 0 to any number of characters; default value 1. The format of the Version Number is a Major Version Number followed by an optional period and an optional Minor Version Number. Major and Minor Version Numbers have a range of 0 through 255. The following are examples of Version Numbers: 1, 1.0, 2.12, 255.255. The following Version Numbers are equivalent: 1, 1., 1.0, 1.00. If the Version Number is equal to the version supported by the ADVIA Centaur CP system, the current incoming message is accepted for processing. If the Version Number is greater than the version supported by the ADVIA Centaur CP system, the current incoming message is ignored.

The following figure shows an example of a header record.

H|^&| |ACCP1| | | |Host| |P|1<CR>

The following table identifies the values in each field in the above example.

Field	Field Name	Value
H1	Record Type	H (Header)
H2	Delimiter Characters:	
	Field Delimiter	
	Repeat Delimiter	\
	Component Delimiter	^
	Escape Delimiter	&
H5	Sender ID	ACCP1
H10	Receiver ID	Host
H12	Processing ID	P (Production)
H13	Version Number	1

Patient Record

The patient record is used to supply patient demographic information for order and result records that follow it. Its hierarchical level is 1. The following table describes the values each field can contain.

Field	Field Name	Value
P1	Record Type	Outgoing: 1 character; always P . Incoming: 1 character; can be P or p .
P2	Sequence Number	Outgoing: 1–5 characters. Formatted as a left justified number with a range from 1 through 65535 . Equal to the <i>n</i> th occurrence of a patient record within the current outgoing application message. Used for validating the integrity of the application message by ensuring that the message contains all records. Incoming: 1 to any number of characters, with a default value of 0 , which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the <i>n</i> th occurrence of a patient record within the current incoming application message. If the current Sequence Number does not equal the Sequence Number of the expected record, an error is reported.

Field	Field Name	Value
P3	Practice Assigned Patient ID	Outgoing: 0–11 characters. Set to the Patient ID (PID) field of an outgoing ADVIA Centaur CP system worklist or result entry. Incoming: 0–11 characters. Written to the PID field of imported worklist entries. An empty field causes no change to the database field. A value of two double quotes ("") deletes the database field. This field is used only for user information. The ADVIA Centaur CP system worklist allows the same PID field to exist in multiple worklist entries.
P6	Patient Name	Outgoing: 0–3 components. Total length of the field is 0–30 characters. Set to the patient Name field of the worklist entry or result. Each name component is separated by the Component Delimiter character (^), for example, Smith^John^A. Trailing empty name components and trailing Component Delimiter characters are not included. For example, Patient Name is set to Smith for a patient worklist entry with the last name Smith. Incoming: 0–3 components. (Components 4 and up are ignored.) Written to the patient Name field of the worklist entry or result. Total length of the field is 0–30 characters. An empty field causes no change to the database field. A value of two double quotes ("") causes the database field to be deleted.
P6.1	Last Name	0–30 characters.
P6.2	First Name	Maximum length dependent on the length of the Last Name.
P6.3	Middle Initial	Maximum length dependent on the length of the Last and the First Names.

Field	Field Name	Value
P8	Date of Birth (DOB)	Outgoing: 0 or 8 characters. Set to ANSI X3.30 format of patient date of birth (DOB) in worklist entry or result, for example, 19721005 for October 5, 1972. If the patient DOB field is not specified in the worklist or result entry, this field is empty. The allowed range is 10010101 through 210012310000 (January 1, 1001 through December 31, 2100). Incoming: 0 or 8 characters representing ANSI X3.30 date format. Written to date of birth (DOB) field in the worklist entry. An empty field causes no change to the database field. A value of two double quotes (") deletes the database field. Invalid dates are treated as an empty field. This field is used for establishing normal ranges for the sample.
P9	Patient Sex	Outgoing: 1 character. Value set by the Patient Sex field in worklist entry or result. If sex is male, field is set to M. If sex is female, field is set to F. If the sex is unknown, the field is set to U. If the sample type is a control, field is blank. Incoming: 0 or 1 character. Written to the Patient Sex field in worklist entry. If Patient Sex value is M, database field is set to male. If Patient Sex value is F, database field is set to female. If Patient Sex value is U, database field is set to unknown. An empty field causes no change to the database field. A value of two double quotes (") causes the database field to be set to unknown sex. An invalid Patient Sex value causes the database field to be set to unknown sex. This field is used for establishing normal ranges for the sample.
P14	Attending Physician ID	Outgoing: 0–12 characters. Set to the Physician field of worklist or result entry. Incoming: 0–12 characters. Written to the Physician field of imported worklist entry. An empty field causes no change to the database field. A value of two double quotes (") deletes the database field. This field is used only for user information.

Field	Field Name	Value
P26	Location	Outgoing: 0–12 characters. Set to the Location field of worklist entry. Incoming: 0–12 characters. Written to the Location field of imported worklist entry. An empty field causes no change to the database field. A value of two double quotes ("") deletes the database. This field is for user information only.

The following figure shows an example of a patient record.

```
P|12|A123-45-6789| | |Jones^Mary^A| |19540601|F| | | |12B| | | | | | | | |
|ER1<CR>
```

The following table identifies the values in each field in the above example.

Field	Field Name	Value
P1	Record Type	P (Patient)
P2	Sequence Number	12
P3	Practice Assigned Patient ID	A123-45-6789
P6	Patient Name	
P6.1	Last Name	Jones
P6.2	First Name	Mary
P6.3	Middle Initial	A
P8	DOB	June 1, 1954
P9	Patient Sex	F (Female)
P14	Attending Physician ID	12B
P26	Location	ER1

Patient Comment Record

The patient comment record is an optional record used to supply additional patient demographic information in free text. It is present if the Comment field in the worklist database is not null. Its hierarchical level is 2. The patient comment record always follows the patient record.

The ADVIA Centaur CP system retains up to three patient comments. If the system sends more than three comments, the system retains only the last three comments. The ADVIA Centaur CP system transmits only the comments that contain non-null text. The following table describes the allowable values for each field.

NOTE: The system also uses comment records as result comment records or as general comment records associated with other record types. Refer to *Result Comment Record*, page 3-38, *Error Detection and Recovery*, page 3-8, for more information.

Field	Field Name	Value
C1	Record Type	Outgoing: 1 character; always C . Incoming: 1 character; can be C or c .
C2	Sequence Number	Outgoing: 1 character, in the range from 1 through 3 . If a patient comment record is needed, the ADVIA Centaur CP system transmits up to a maximum of three patient comment records per patient record. Used for validating the integrity of the application message by ensuring that the message contains all records. Incoming: 1 to any number of characters, with a default value of 0 , which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the <i>n</i> th occurrence of a patient comment record immediately following the last patient record, which is considered the current patient record. If the current Sequence Number does not equal the Sequence Number of the expected record, a fatal parse error is reported.
C3	Comment Source	Outgoing: 0 characters; always empty. Incoming: not used for incoming message processing.
C4	Comment	Outgoing: 2 components. Set to the Comment field of the patient sample or worklist entry. Incoming: 1 component. Set to the Comment field of the worklist.
C4.1	Comment Code	Outgoing: always empty. Incoming: 0–120 characters. Written to the Comment field of the imported worklist entry. An empty Comment Text field or the lack of a patient comment record causes no change to the database field. A value of two double quotes ("") deletes the database field. This field is for user information only.

Field	Field Name	Value
C4.2	Comment Text	Outgoing: 1–120 characters. Set to the Comment field of the outgoing result entry. If the Comment field is empty, a patient comment record is not generated for transmission. Incoming: not used for incoming message processing.
C5	Comment Type	Outgoing: 1 character, G (Generic). Incoming: 0 or 1 character. If the Comment Type field is empty or G (Generic), the comment record is processed as a patient comment record. If the Comment Type field is any other value, the comment record is processed as a general comment record and is ignored by the ADVIA Centaur CP system.

The following figure shows an example of an incoming patient comment record.

C|1| |SPECIMEN HEMOLYZED|G<CR>

The following table interprets the values in each field in the above example.

Field	Field Name	Value
C1	Record Type	C (Comment)
C2	Sequence Number	1
C3	Comment Source	empty
C4	Comment	
C4.1	Comment Code	SPECIMEN HEMOLYZED
C4.2	Comment text	empty
C5	Comment Type	G (Generic)

Order Record

The order record is used to supply information on a specific sample's test requests. This information is necessary either for ordering tests on a specific sample or for reporting results for tests on a specific sample. An order record is associated with the previous patient record. Its hierarchical level is 2.

The following table describes the allowable values for each field.

Field	Field Name	Value
O1	Record Type	Outgoing: 1 character; always O . Incoming: 1 character; can be O or o .

Field	Field Name	Value
O2	Sequence Number	Outgoing: 1 to any number of characters; usually 1. Used for validating the integrity of the application message by ensuring that the message contains all records. Incoming: 1 to any number of characters, with a default value of 0, which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the <i>n</i>th occurrence of an order record since the last patient record, which is considered the current patient record. If the current Sequence Number does not equal the Sequence Number of the expected record, an error is reported.
O3	Specimen ID	Outgoing: 1 - 13 characters. Set to the Sample SID (SID) field of a result entry. Incoming: 1-13 characters. Written to the Sample ID (SID) field of the imported worklist entry.

Field	Field Name	Value
O5	Universal Test ID	Outgoing: List of tests, where each test is described by a multi-component field, separated by a Component Delimiter character (^). Each group of components is separated by a Repeat Delimiter character (!). If the current order record is for submitting a worklist request for a specific sample, this field describes the list of requested tests. If the current order record is for reporting single or multiple result values, this field describes the list of tests whose result values are reported in subsequent result records. The list of tests can contain from one up to the maximum number of tests the ADVIA Centaur CP system can support including ratio and off-system tests. Refer to individual components (field components O5.1 through O5.6) for data format information. Incoming: List of tests, where each test is described by a multi-component field, separated by a Component Delimiter character (^). Each group of components is separated by a Repeat Delimiter character (!). If a test is unknown to the ADVIA Centaur CP system, an error message is returned to the LIS and processing continues on the list of tests. The list of tests can contain any number of names. If a single Universal Test ID matches a ratio test, all tests associated with the ratio are automatically added to the list of tests. Refer to field O12, Action Code, for processing of updates and cancellations of the tests specified in this field. Refer to individual components (field components O5.1 through O5.6) for data format information.
O5.1	Universal Test ID	Outgoing: 0 characters; always empty. Incoming: 0 to any number of characters. This component is always ignored.
O5.2	Universal Test ID Name	Outgoing: 0 characters; always empty. Incoming: 0 to any number of characters. This component is always ignored.
O5.3	Universal Test ID Type	Outgoing: 0 characters; always empty. Incoming: 0 to any number of characters. This component is always ignored.

Field	Field Name	Value
O5.4	Manufacturer's Code	Outgoing: 1–8 case-sensitive characters. The defined LIS code name is used for the test scheduled for the sample. Incoming: 1–8 case-sensitive characters. If a value exists, this value is used to search the ADVIA Centaur CP assay database. The ADVIA Centaur CP system searches the assay database for the matching LIS code. If a match is found, that test is selected for the sample. If no match is found, an error is reported.
O5.5	Dilution Protocol (optional)	Outgoing: empty field Incoming: case-sensitive characters. Use the following dilution options: • neat only • neat and diluted • dilute only • dilution profile
O5.6	Dilution Ratio (optional)	Outgoing: 2, 5, 10, 20, 50, 100, 200, 500, 1000, or 2500 Incoming: Dilution ratio must be defined for the test on the ADVIA Centaur CP. 2, 5, 10, 20, 50, 100, 200, 500, 1000, or 2500, if that dilution ratio is not defined on the ADVIA Centaur CP, then the order is rejected. or enter the name of an ADVIA Centaur CP predefined dilution profile. If that profile is not defined on the ADVIA Centaur CP, then the order is rejected.
O6	Priority	Outgoing: 1 character. Value determined by the stat flag in the worklist entry. If the stat flag is Yes, the Priority field is set to S (Stat). If the stat flag is No, the Priority field is set to R (Routine). Incoming: 1 character. Written to the stat flag in the worklist entry. If the Priority is S, the stat flag is set to Yes. Otherwise, the stat flag is set to No.

Field	Field Name	Value
O12	Action Code	Outgoing: 0 or 1 character. If the worklist entry or the result entry is for a control sample, the Action Code field is set to Q (QC Test Specimen). If the worklist entry or the result entry is for a patient sample, the Action Code field is empty. Incoming: 1 character. If the order record is for an incoming worklist entry (refer to field O26, Report Type), this field is processed as follows: If one of the Action Codes is C (Cancel Request), the tests specified in Universal Test ID field O5 are canceled for the sample specified in field O3. If the Action Code does not specify Cancel Request, the Action Code is interpreted as a New Request or an Add Request, depending upon the existence of an entry in the ADVIA Centaur CP system worklist for the Sample ID from field O3.1. Empty and invalid Action Code fields are interpreted as a New Request or as an Add Request, depending upon the existence of an entry in the ADVIA Centaur CP system worklist for the Sample ID from field O3.1.

Field	Field Name	Value
O26	Report Type	Outgoing: Zero or more repetitions of 1 character. If the outgoing message contains rejected incoming worklist data, the Report Type field contains X (Work Cannot Be Done). If the outgoing message contains result data, the Report Type field contains F (Final Results). If the outgoing message contains data in response to a query, the Report Type field contains Q (Response to a Query). If the outgoing message is a response to a remote query for pending results, the Report Type field contains I (Instrument Pending). Corrections of test results, are transmitted with an "C" flag. This is the case if a result was changed using the "Edit" function of Details in the Results Tab. Incoming: Zero or more repetitions of 1 character If the test order is not canceled (See Action Code (Field O12)) then the Report Type is assumed to be O (Order). If the test order is canceled (See Action Code (Field O12)) then the Report Type is ignored.

The following figure shows an example of an order record.

```
O|1|18653| |^^T4^^HCG^^P1234|R| | | |Q| | | | | | | | |O\Q<CR>
```

The following figure shows an example of an order record with a dilution profile.

```
O|1|SID1001||^^T4^dilution profile^DilProName|R| | | | | | | | | | | | | | |F<CR>
```

The following figure shows an example of an order record with a dilution.

```
O|1|18653| |^^THCG^dilute only^200|S| | | | | | | | | | | | | | | | | | | |F<CR>
```

The following table interprets the values in each field in the above example.

Field	Field Name	Value
O1	Record Type	O (Order)
O2	Sequence Number	1
O3	Specimen ID	
O3.1	Sample ID	18653
O5.4	Manufacturer's Code	THCG
O5.5	Dilution Protocol	dilute only
O5.6	Dilution Ratio	200
O6	Priority	S (Stat)
O12	Action Code	empty
O26	Report Type	F (Final result)

Result Record

The result record supplies the test result information for a single test of a specific sample. Results of multiple tests are specified in separate result records. The ADVIA Centaur CP system rejects incoming result records from a remote system. The hierarchical level is 3. The result record is associated with the previous order record. The following table describes the allowable values for each field.

Field	Field Name	Value
R1	Record Type	Outgoing: 1 character; always R . Incoming: Not processed.
R2	Sequence Number	Outgoing: 1–5 characters, formatted as a left justified number, with a range from 1 through 65535 . Equal to the <i>n</i> th occurrence of a result record since the last order record. Used to verify the integrity of an application message by ensuring that the message contains all records. Incoming: Not processed.
R3	Universal Test ID	Outgoing: A single test name described by 8 components, where each component is separated by a component delimiter character (^). The Universal Test ID specifies the name of the test associated with the result. Refer to the individual components (field components R3.1 through R3.8) for data format information. Incoming: Not processed.
R3.1	Universal Test ID	Outgoing: 0 characters; always empty.

Field	Field Name	Value
R3.2	Universal Test ID Name	Outgoing: 0 characters; always empty.
R3.3	Universal Test ID Type	Outgoing: 0 characters; always empty.
R3.4	Manufacturer's Code	Outgoing: 1–8 characters. The LIS code of the test.
R3.5	Dilution Protocol (optional)	Outgoing: Empty Field
R3.6	Dilution Ratio (optional)	Outgoing: 2, 5, 10, 20, 50, 100, 200, 500, 1000, or 2500
R3.7	Replicate Number (optional)	Outgoing: Ignored. Only the mean of the replicates are transmitted, if the assay has Average Replicates enabled. If not enabled then each replicate is transmitted as an individual test result.
R3.8	Result Aspects	Outgoing: 0–4 characters. One of 4 possible text strings used to identify the specific aspect of the result being transmitted. The following are possible values and their interpretations: • DOSE - Result concentration value (in Master Curve units) • INDX - Result index value • INTR - Result interpretation • RLU - RLU value used in result calculation Incoming: Not processed
R4	Data Value	Outgoing: 1–15 characters. Depending on the result aspect, several result formats are available at the Data Value field: • number - concentration value • qualitative - text string • alphanumeric - < value • alphanumeric - > value Incoming: Not processed.

Field	Field Name	Value
R5	Units	Outgoing: 0–6 characters. If the Data Value field (R4) represents a quantitative result, this field is set to the Units field of the definition of the associated test. If the Data Value field represents a qualitative result, the Units field is empty. Incoming: Not processed.
R7	Result Abnormal Flags	Outgoing: Zero or more repetitions of 1 character. The repetitions represent some of the ADVIA Centaur CP system result flags and are one of the following: If the result is flagged as low, this field contains L (Below Reference Range). If the result is flagged as high, this field contains H (Above Reference Range). If the result is flagged as under range, this field contains < (Below Concentration Range). If the result is flagged as over range, this field contains > (Above Concentration Range). These flags can occur in combination, separated by repeat delimiters (I), if the test has multiple replicates with widely varying results. Refer to <i>Result Comment Record</i>, page 3-38, and <i>Error Detection and Recovery</i>, page 3-8 for other ADVIA Centaur CP system result flags. Incoming: Not processed.
R9	Result Status	Outgoing: One or more repetitions of 1 character. The Result Status field has a value of F (Final) or P (Preliminary). Other values can occur, separated by the repeat delimiter (I). If the result is sent in response to a query, the field has a value of Q. If a previously sent result was changed by the system operator and not sent after the change, the result status field has a value of C to indicate correction of a previously transmitted result. If a result was sent previously, the result status field has a value of R. Values of R and C do not occur in combination. If the outgoing message contains rejected incoming worklist data, the Report Type field contains X (Work Cannot Be Done). Incoming: Not processed.

Field	Field Name	Value
R13	Date Test Completed	Outgoing: 12 characters. Set to ANSI X3.43 format of date and time values. The ADVIA Centaur CP system transmits a 12 character string in YYYYMMDDHHMM form. Hours are in military (24-hour) form. For example, 199808051403 stands for August 5, 1998, 14:03 (2:03 p.m.). Incoming: Not processed.

The following figure shows an example of a result record.

```
R|1|^^^FER^^^DOSE|45.0|ng/mL| |L| |F\Q\R| | |199806291534<CR>
```

The following table interprets each field in the above example.

Field	Field Name	Value
R1	Record Type	R (Result)
R2	Sequence Number	1
R3	Universal Test ID	
R3.4	Manufacturer's Code	FER (Bayer HealthCare LIS code for the ADVIA Centaur CP Ferritin assay)
R3.8	Result Aspects	DOSE
R4	Data Value	45.0
R5	Units	ng/mL (nanograms per milliliter)
R7	Result Abnormal Flags	L (Low, below reference range)
R9	Result Status	F (Final result) Q (Sent in response to a query) R (Result previously transmitted)
R13	Date Test Completed	June 29, 1998, 15:34 (3:34 p.m.)

Result Comment Record

The result comment record is used to communicate additional flags beyond those supported by the Result Abnormal Flags field of the result record. The record is optional. It is used if a flag other than high (**H**), low (**L**), over range (**>**), or under range (**<**) is associated with the result. Each flag generates an additional result comment record. The ADVIA Centaur CP system ignores result data from a remote system. The system rejects incoming result comment records. Its hierarchical level is 4. The result comment record always follows a result record.

The ADVIA Centaur CP system retains up to three result comments entered by the operator. The system transmits each comment in its own result comment record. The following table describes the allowable values for each field.

NOTE: The system also uses comment records as patient comment records or as general comment records associated with other record types. Refer to *Patient Comment Record*, page 3-26, and *Error Detection and Recovery*, page 3-8, for more information.

Field	Field Name	Value
C1	Record Type	Outgoing: 1 character; always C . Incoming: Not processed.
C2	Sequence Number	Outgoing: 1–5 characters, formatted as a left justified number, with a range from 1 through 65535 . Equal to the <i>n</i> th occurrence of a result comment record within the current outgoing application message. Used to verify the integrity of an application message by ensuring that the message contains all records. Incoming: Not processed.
C3	Comment Source	Outgoing: 0 or 1 character. If the comment is a result flag, the Comment Source field is set to I . If the comment is a operator-entered comment, this component is blank. Incoming: Not processed.
C4	Comment	Outgoing: 2 components. Set to the Comment field of the test result record. Incoming: Not processed.
C4.1	Comment Code	Outgoing: 0–60 characters. Result flags map to the values in the following table. If the flags do not exist for the result record, result comment records are not generated for transmission. If the comment is a operator-entered result comment, this component is blank. Incoming: Not processed
C4.2	Comment Text	Outgoing: 1–120 characters. The text for any operator-entered comments. If the comment is an internal result flag, this component is blank. Incoming: Not processed.
C5	Comment Type	Outgoing: 1 character; either G or I . If there is a result flag, this component is I . If there is a operator-entered comment, this component is G . Incoming: Not processed.

The following shows the text transmitted in the comment code field of the result comment record.

ADVIA Centaur CP flags sent in the outgoing Comment Field C4.1

- < Check Range
- < Linearity
- < Low Range
- > Check Range
- > Dilution Pt
- > High Range
- > Linearity
- Autorepeat
- Cal Accepted
- Cal Assigned
- Cal expired
- Cal invalid
- Diluted
- Edited
- High CV
- In Range
- Incomplete
- Investigational
- Low Signal
- Manual Dilution
- No calculation
- No Mastercurve
- No Mitigation
- Overdiluted
- Reagent expired
- Repeat
- Repeated
- Replicate excluded
- Signal 1
- Signal 3
- Signal 4
- Signal 5
- Waived

The following sample statuses have no corresponding result comment text:

- Pending
- To do
- Placed

The following figure shows an example of a result comment record.

C[1]I|< Check Range|I<CR>

The following table interprets the values in each field in the above example.

Field	Field Name	Value
C1	Record Type	C (Comment)
C2	Sequence Number	1
C3	Comment Source	I (Instrument)
C4	Comment	
C4.1	Comment Code	Above Check
C4.2	Comment Text	empty
C5	Comment Type	I (Instrument flag)

Query Record

The query record is used to request data from remote systems. Its hierarchical level is 1. The ADVIA Centaur CP system processes only one received query at a time. If a second query is received while the first is processing, the system cancels both queries. Refer to the description of field Q13 for proper cancellation of queries.

The following table describes the allowable values for each field.

Field	Field Name	Value
Q1	Record Type	Outgoing: 1 character; always Q . Incoming: 1 character; can be Q or q .
Q2	Sequence Number	Outgoing: 1 character; always 1 . The ADVIA Centaur CP system always ends an application message before transmitting another query record. Used to verify the integrity of an application message by ensuring that the message contains all records. Incoming: 1 to any number of characters, with a default value of 0 , which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. Equal to the <i>n</i> th occurrence of a query record within the current incoming application message. If the current Sequence Number does not equal the Sequence Number of the expected record, an error is reported.

Field	Field Name	Value
Q3	Starting Range ID Number	Outgoing: 0, or 2 components. If the query is a query cancellation of an outstanding query, this field is empty. If the operator requested a worklist, Sample ID, has no restrictions and the field is set to ALL. If the ADVIA Centaur CP system has initiated the worklist requests automatically, the field has 2 or 4 components. The first is the Patient ID, which is always empty and separated by the component delimiter (^) from the Sample ID. Refer to the individual components (field components Q3.1 and Q3.2) for data format information. Incoming: 0, or 2 components. If the field is empty, invalid, or ALL, no restrictions are made on Patient ID or Sample ID. Otherwise, the field has 1, 2, or 4 components, separated by the component delimiter (^) and is a single value or a list of ranges. The system assumes the Starting Range ID and an Ending Range ID define a range. You can specify ID ranges by specifying a Starting Range ID and an Ending Range ID, separated by the repeat delimiter. For example, the first Starting Range ID matches with the first Ending Range ID. If a value is present in the Starting Range ID field, without a matching Ending Range ID field, an open-ended range is assumed. The range includes all specimen ID's greater than the specified Starting Range ID. If a value is present in the Ending Range ID field without a matching Starting Range ID field, an open-ended range is assumed. The range includes all specimen IDs less than that specific Ending ID. Multiple ranges are specified by using the repeat delimiter. The entries selected meet a logical AND of all the selected criteria. Refer to the individual components (field components Q3.1 and Q3.2) for data format information.
Q3.1	Patient ID	Outgoing: 0 characters. Incoming: 0–11 characters. The Patient ID is used as a criterion for matching the PID field in result entries.

Field	Field Name	Value
Q3.2	Sample ID	Outgoing: 1–13 characters. The Sample ID is selected from the worklist SID field. Incoming: 0–13 characters. The Sample ID is used as a criterion on matching the SID field in result entries.
Q4	Ending Range ID Number	Outgoing: This field is always empty for all ADVIA Centaur CP system-initiated query records. Incoming: 0, or 2 components. If the field is empty or invalid, the Starting Range ID Number field is treated as a single value, provided that value is present in the Starting Range ID Number field. If both the Starting Range ID Number field and the Ending Range ID Number field are empty or invalid, no restrictions are made on Patient ID or Sample ID. If the Starting Range ID Number field has a single value, the Ending Range ID Number field is an end range value with up to two components, separated by the component delimiter (^), up to the last non-empty component. Refer to the individual components (field components Q4.1 and Q4.2) for data format information.
Q4.1	Patient ID	Outgoing: 0 characters. Incoming: 0–13 characters. The Patient ID is used as a criterion for matching the PID field in worklist or result entries.
Q4.2	Sample ID	Outgoing: 0 characters. Incoming: 0–13 characters. The Sample ID is used as a criterion for matching the SID field in worklist entries or results.

Field	Field Name	Value
Q5	Universal Test ID	Outgoing: 0 or 1 component. If the record is for a query cancellation message, this field is empty. If the operator requests a worklist, no restrictions are made on test IDs and the field is set to ALL. If the ADVIA Centaur CP system initiates the worklist requests automatically, the field is set to ALL. Incoming: 0, 1, or 4 components. If empty, invalid, or ALL, there are no restrictions of record selection based on test. Otherwise, the field is a single value or list of test names. Each test is 4 components and each group of 4 components is separated by a repeat delimiter character. If a name is unknown to the ADVIA Centaur CP system, the test name is ignored and processing continues on the list of test names. The list of test names can contain any number of names. Refer to the individual components (field components Q5.1 through Q5.4) for data format information.
Q5.1	Universal Test ID	Outgoing: 0 characters; always empty. Incoming: 0 to any number of characters. This component is always ignored.
Q5.2	Universal Test ID Name	Outgoing: 0 characters; always empty. Incoming: 0 to any number of characters. This component is always ignored.
Q5.3	Universal Test ID Type	Outgoing: 0 characters; always empty. Incoming: 0 to any number of characters. This component is always ignored.
Q5.4	Manufacturer's Code	Outgoing: 0 characters; always empty. Incoming: 0–8 characters. If a value exists, this value is used to search the ADVIA Centaur CP assay database. The ADVIA Centaur CP system searches the test definition database for the matching LIS code. If no match is found, the system searches for the matching test name.
Q6	Nature of Request Time Limits	Outgoing: 0 characters. Incoming: 0 or 1 character. If blank or R, the dates and times in Q7 and Q8 are used to match result entries. If invalid or S, the dates and times in Q7 and Q8 are ignored.

Field	Field Name	Value
Q7	Beginning Request Results Date	Outgoing: 0 characters. Incoming: 0–14 characters. This field is used as a criterion for matching the date and time a test was completed in result entries. Set to ANSI X3.43 format of date and time values. This is up to a 14-character string in YYYYMMDDHHMMSS form. The ADVIA Centaur CP ignores seconds. Hours are in military (24-hour) form. The ADVIA Centaur CP system matches the portions of the date and time supplied. Valid examples: 199807 (YYYYMM) matches all result entries completed in July of 1998; 1998080123 (YYYYMMDDHH) matches all result entries completed in the twenty-third hour (11:00 to 11:59 p.m.) of August 1, 1998. Invalid examples: 199823 (YYYYHH) and 1998 01 (YYYY DD). A Beginning Request Results Date and an Ending Request Results Date define a range. If a value is present in the Beginning Request Results Date field without a matching Ending Request Results Date, an open-ended date range is assumed. The range includes all results with a date greater than the Beginning Request Results Date. If a value is present in the Ending Request Results Date field without a matching Beginning Request Results Date, an open-ended date range is assumed. The range includes all results with a date less than the Ending Request Results Date. You can specify date ranges by specifying a Beginning Request Results Date and an Ending Request Results Date, separated by the repeat delimiter. For example, the first Beginning Request Results Date matches with the first Ending Request Results Date.

Field	Field Name	Value
Q8	Ending Request Results Date	Outgoing: 0 characters. Incoming: 0–14 characters. This field is used as a criterion for matching the date and time a test was completed in result entries. Set to ANSI X3.43 format of date and time values. The value of the field is processed similarly to the beginning request results date and time. The field is empty or contains a single date and time.
Q13	Request Information Status Code	Outgoing: 1 character. For an operator request for work or for an system-initiated automatic request for work, the value of the request information status code is O , which is a request for tests and demographics only. Incoming: Not p[rocessed.

The following figure shows an example of an ADVIA Centaur CP system-initiated worklist request query record.

Q1|^SID12-A|^SID12-A |ALL| ||||| |O<CR>

The following table interprets the values in each field in the above example.

Field	Field Name	Value
Q1	Record Type	Q (Query)
Q2	Sequence Number	1
Q3	Starting Range ID Number	
Q3.1	Patient ID	empty (no restrictions)
Q3.2	Sample ID	SID12-A (starting value of the range)
Q4	Ending Range ID Number	
Q4.2	Sample ID	SID12-A (ending value of the range)
Q5	Universal Test ID	ALL (request all tests; no restrictions)
Q7	Beginning Request Results Date	empty (no restrictions)
Q8	Ending Request Results Date	empty (no restrictions)
Q13	Request Information Status Code	O (request tests)

The following figure shows an example of an operator-initiated worklist request query record.

Q1|ALL|ALL| || || ||O<CR>

The following table interprets the values in each field in the above example.

Field	Field Name	Value
Q1	Record Type	Q (Query)
Q2	Sequence Number	1
Q3	Starting Range ID Number	ALL (request all samples; no restrictions)
Q4	Ending Range	empty (no restrictions)
Q5	Universal Test ID	ALL (request all tests; no restrictions)
Q7	Beginning Request Results Date	empty (no restrictions)
Q8	Ending Request Results Date	empty (no restrictions)
Q13	Request Information Status Code	O (request tests)

The following figure shows an example of a worklist request query record that cancels any outstanding queries.

Q1| || || || || ||A<CR>

The following table interprets the values in each field in the above example.

Field	Field Name	Value
Q1	Record Type	Q (Query)
Q2	Sequence Number	1
Q3	Starting Range ID Number	empty (no restrictions)
Q4	Ending Range ID Number	empty (no restrictions)
Q5	Universal Test ID	empty (no restrictions)
Q7	Beginning Request Results Date	empty (no restrictions)
Q8	Ending Request Results Date	empty (no restrictions)
Q13	Request Information Status Code	A (cancel previously transmitted query)

Termination Record

The termination record is the last record of a message and closes the message. It provides an explanation for ending the message. Its hierarchical level is 0. The following table describes the allowable values for each field.

Field	Field Name	Value
L1	Record Type	Outgoing: 1 character; always L . Incoming: 1 character; can be L or I .
L2	Sequence Number	Outgoing: 1 character; always 1 . Incoming: 1 to any number of characters, with a default value of 0 , which is always an invalid Sequence Number. Formatted as a string of ASCII numeric characters. If the current Sequence Number does not equal the Sequence Number of the expected record, a fatal parse error is reported. This value should always be 1 because multiple termination records not separated by a header record generate a fatal parse error.

Field	Field Name	Value
L3	Termination Code	Outgoing: 0 or 1 character. If the ending of the message is normal, this field is absent or N (Normal). A Termination Code of F (Final) is used in reply to a query to indicate that all available information is sent in response. A Termination Code of I . (No Information Available) is used in reply to a query if no information is available to send in response to the query. A Termination Code of Q (Query in Error) is used in reply to a query when another query is still outstanding or the Request Information Status Code of the query record is invalid. After sending a termination record with a Termination Code of F , I , or Q , the ADVIA Centaur CP system considers all external queries closed and is ready to process new queries.

Field	Field Name	Value
L3	Termination Code	Incoming: 0 or 1 character, with a default value of N (normal). If the Termination Code is null or N, the system closes the incoming message and logs the message as normally terminated. If the value is T, the system closes the incoming message and logs the message as canceled by the sender. If the value is R, the system closes the incoming message and logs the message as canceled by the receiver. If the termination code is F (Final), the system closes the incoming message and logs the message as the end of the query response. The current query on the port is closed out, which allows any pending query to transmit. If the code is I, the system closes the incoming message and logs the message as having no information found in response to the current query. The current query on the port is closed out, which allows any pending query to transmit. If the code is Q, the system closes the incoming message and logs the message as stating that the current query that was sent by the ADVIA Centaur CP system to the remote system was in error. The current query on the port is closed out, which allows any pending query to transmit. The system handles all other character values as having an unknown message termination status. The incoming message is closed out, but no message closing status is logged.

The following figure shows an example of a termination record.

L|1|F<CR>

The following table interprets the values in each field in the above example.

Field	Field Name	Value
L1	Record Type	L (Termination)
L2	Sequence Number	1
L3	Termination Code	F (final record of query response)

Data Link Layer Interaction

Outgoing application messages are sent to the data link layer on a message basis. Since a message consists of a number of records, the application layer sends all of the records for a message together. This allows the data link layer to block data in the most efficient manner.

There is no connection between the application messages and the link establishment phase and the link release phase of the physical connection. The data link layer can release the link in the middle of an application message, and send multiple application messages during a single session. The operator cannot influence when data link sessions terminate.

The application layer is solely responsible for parsing the messages from the incoming stream. The application layer views incoming data as a character stream.

ASTM Protocol Exceptions

The ADVIA Centaur CP system implementation of the ASTM protocol deviates from the ASTM standard in several minor ways:

- The practice fields in the patient record and the user fields in the order record are not part of the instrument database. The system does not record or return these fields with results.
- The system does not support highlight sequences, unhighlight sequences, and local (manufacturer) escape sequences. The system receives these escape sequences, ignores them, and does not transmit them.
- The ADVIA Centaur CP system accepts null application records, which are ignored, letting processing continue normally.

The system uses and recognizes all other predefined sequences during transmission.

Query Timer

The sender uses the query timer to limit the amount of time to wait for a reply to a query. If the wait is longer than the timeout specified, a query timeout occurs and the sender cancels the outstanding query. ASTM protocol allows only one outstanding query at a time.

Error Handling

The ADVIA Centaur CP system communications software uses the Event Log screen to report errors. Refer to Section 4, *Communications*, for more information.

The communications software handles special situations during in the course of communicating with a remote system. These situations include failure to process incoming records, transmission errors, abnormal message termination codes, unrecognized message record sequences (and other parse errors), and a query cancellation.

The ADVIA Centaur CP system attempts to recover from most input message errors. The ADVIA Centaur CP system may reject an erroneous query and test order partially and totally based on the level of security of the detected error. For example, an error in one specimen identifier of an order record causes the system to reject all tests ordered on that specimen. In contrast, an invalid test identifier causes the system to reject only that test order. A very severe error such as an invalid header record in an order message causes the system to reject the whole message. A fatal error in a query record causes the system to reject the entire query. As for order errors, if the effect of the error is localized, the system only rejects the affected test requests.

The ADVIA Centaur CP system communications software prepares a reply message to send to the remote system. The ADVIA Centaur CP interface returns as much data as it can salvage of every erroneous item that it rejects to the LIS host. It consists of a header, a patient record containing the rejected data, an order record containing the rejected data with a report type of **X** (cannot process), a comment record describing the reason for the rejection, and a termination record. The exception to this rule is the case of unrecognized messages. For these, the error messages consist of a header record, a comment record, and a termination record. ASTM provides no error codes for this case. LIS hosts have only the contents of the comment record to process the error.

The following table describes the reasons the ADVIA Centaur CP system rejects an incoming record.

Reason	Description
invalid sample ID	An order record has a sample ID of "" or the sample ID is absent.
result record received	An order record with a following result record.
bad dilution profile	The dilution profile is not predefined in the ADVIA Centaur CP system.
bad dilution protocol	The sent dilution factor does not match the list of possible dilution factors for that test.
bad SID	The SID does not conform to the minimum character length range of 4 to 8 digits or the maximum character length range of 11 to 17 digits.
bad test name	Test name does not exist or no profile exists for that test
SID is for control or calibrator	Worklist request for control or calibrators must be ordered at the ADVIA Centaur CP.
multiple workorder	Test request for a pending test.
Wrong LIS version number	Header record for the LIS version is incorrect.
Internal LIS error	Communication error.
Unknown error	Unassigned error.

A transmission error occurs when the data link layer driver reports that it is unsuccessful in sending a record to the remote system. The application may attempt additional transmissions.

Two classes of abnormal termination codes are possible in a received termination record. The first class (with a termination code of **T**, **R**, or **E**) indicates an error in data processing. The current incoming message is closed and the received data is processed according to standard rules. The second class of abnormal terminations relates to responses to queries. A **Q** code indicates a query was sent that the receiver cannot process. An **I** code indicates that no data was found in response to a query. An **F** code is a normal termination of a response to a query indicating that all data is sent.

A fatal parse error in a record is treated like an abnormal termination. The current message is closed and any correctly received data is processed according to standard rules.

If the remote system sends a query message with a cancel status code, it cancels the reply in process in the following manner. If the system does not send a response, the remote system cancels it. If the system sends a response (no reply in process), the system ignores the cancellation. If the system is currently sending a reply, the remote system stops it and sends a termination record with a termination code of **R**.

Depending on the query mode selected, the ASTM application layer only permits one outstanding query for a system. If the remote system sends a second query message to the ADVIA Centaur CP system before the first one is complete, the system automatically cancels the first query. The system rejects the second query and responds to it as an invalid query. The system sends a message containing a header record and a termination record with a termination code of **Q**. This resets pending query states on both the ADVIA Centaur CP system side and the remote system side.

4 Communication

The ADVIA Centaur CP system event log provides information about communication errors and accesses information about possible causes and corrective actions.

The Field Service Engineer (FSE) or LIS contact person can also monitor the incoming and the outgoing messages of the data link layer and the application layer by accessing Setup, LIS Tab, Logfile frame and turning on one or both Record ASTM 1381 and Record ASATM 1394. Refer to the *ADVIA Centaur CP Operator's Guide* and Online Help on how to access the various screens from the workspace.

Using the windows explorer, access D:\Melias\data\Temp\LIS and open either C3R0000*.TRC or C3B0000*.TRC in notepad to review the ASTM 1381 and 1394 logs.

```
2004-12-21, 18:11:27,062-> H|\&||||Host||||ACCP1||P|1<CR>
2004-12-21, 18:11:27,406-> P|1|P1234567038|||Tim Tam||1968909|M||||MD_DOC<CR>
2004-12-21, 18:11:27,843-> O|1|LIS3023||^T4^TSH^TUP|R||||||||O<CR>
2004-12-21, 18:11:28,156-> L|1|N<CR>
2004-12-21, 18:20:18,718-> H|\&||||Host||||ACCP1||P|1<CR>
2004-12-21, 18:20:19,281-> P|1|P1234567039|||Trish Trane||1968909|F||||MD_DOC<CR>
2004-12-21, 18:20:19,718-> O|1|K3024||^T4^TSH^TUP|R||||||||O<CR>
2004-12-21, 18:20:19,906-> L|1|N<CR>
```

1 Data from the C3R0000*.TRC file.

```
2004-12-21, 18:11:26,765-> <ENQ>
2004-12-21, 18:11:26,843<- <ACK>
2004-12-21, 18:11:26,875-> <STX>1H|\&||||Host||||ACCP1||P|1<CR><ETX>EE<CR><LF>
2004-12-21, 18:11:27,093<- <ACK>
2004-12-21, 18:11:27,125-> <STX>2P|1|P1234567038|||Tim Tam||1968909|M||||MD_DOC<CR><ETX>5F<CR><LF>
2004-12-21, 18:11:27,437<- <ACK>
2004-12-21, 18:11:27,468-> <STX>3O|1|LIS3023||^T4^TSH^TUP|R||||||||O<CR><ETX>A6<CR><LF>
2004-12-21, 18:11:27,875<- <ACK>
2004-12-21, 18:11:27,906-> <STX>4L|1|N<CR><ETX>07<CR><LF>
2004-12-21, 18:11:28,187<- <ACK>
```

1 Data from the C3B0000*.TRC file.

When finished troubleshooting, access Setup, LIS Tab, Logfile frame and turn off both Record ASTM 1381 and Record ASATM 1394.

5 Troubleshooting

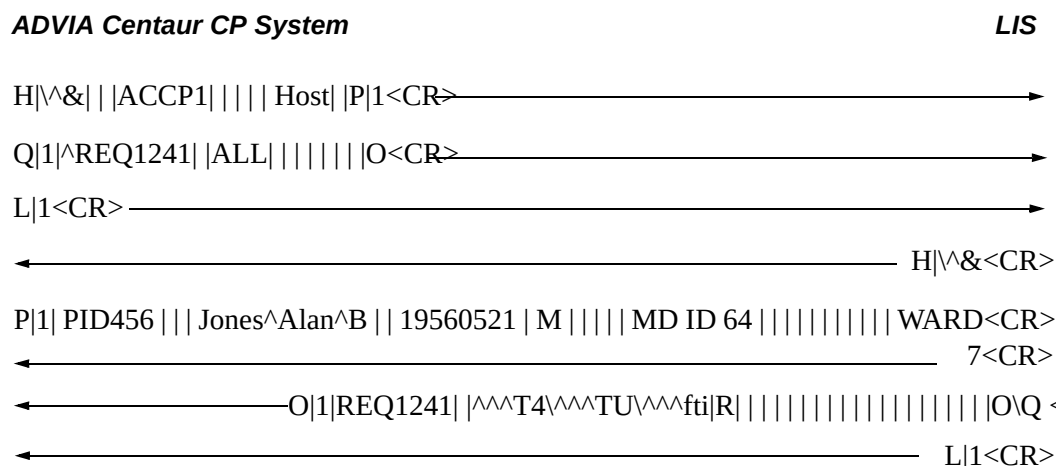
Delays in Automatic Worklist Requests

When using automatic worklist requests, there may be gaps between cuvettes in the ring because each query from the ADVIA Centaur CP system about a sample ID can be separated by 30 or more, depending on the timeout settings in the LIS setup tab.

Possible Cause: Termination Record

Although the remote system responds to the query, it does not close the query. When the last message in response to a query is sent, the termination record should have a value of **F** in the Termination Code field. The ADVIA Centaur CP system waits for a termination record with a value of **F**, **Q**, or **I** in the Termination Code field or waits until the query timer expires before issuing the next query. The Query Timeout field at the LIS tab of the Setup screen is set to 30 seconds.

The following figure gives an example of an incorrect termination record. Note the absence of an **F** value in the Termination Code field in the termination record of the response from the remote system. In this example, if the remote system completes the response to a query from the ADVIA Centaur CP system and the query timeout on the ADVIA Centaur CP system is 30 seconds, then the ADVIA Centaur CP system does not automatically query for the next sample until the 30 seconds elapses, depending on the timeout settings in the LIS setup tab.



Solution

Use a value of **F** in the Termination Code field of the termination record that closes the sequence of messages sent to the ADVIA Centaur CP system in response to each query.

L|1|F<CR>

Possible Cause: LIS Response to the Query

The remote system responds very slowly to each query from the ADVIA Centaur CP system.

Solution

Contact your LIS Administrator to check the parameters at your LIS.

Missing Worklist Response to a Request

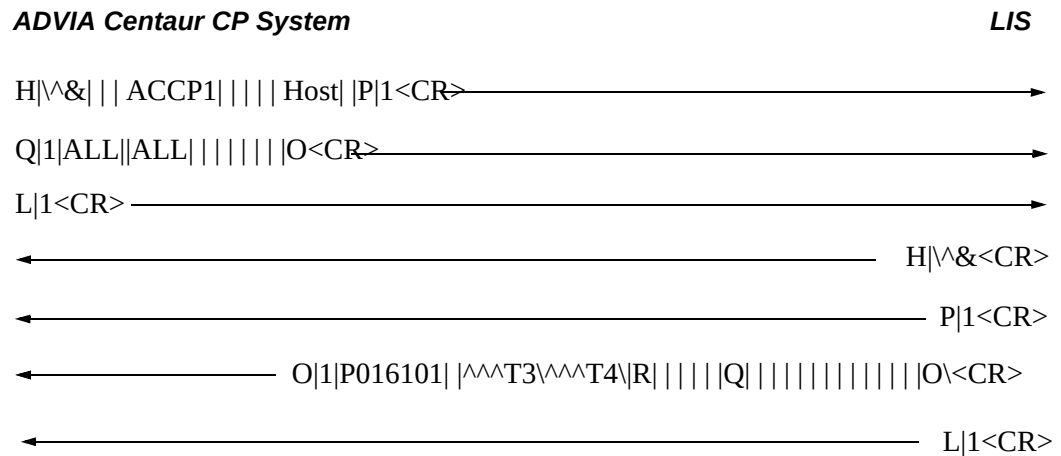
An operator requests a worklist, but the ADVIA Centaur CP system sends a message to the ACCP event log saying the query has timed out and no order record was received for that SID. The ACCP sends the next query request before the query is complete.

Possible Cause: Query Timeout

The query timeout is too short or the remote system does not indicate that order records were sent in response to a query. A value of **OIQ** in the Report Type field of an order record indicates that the record is requesting tests in response to a query. The ADVIA Centaur CP system resets the query timer every time it encounters a value of **Q** in the Report Type field of an order record. The system ignores all received result records. Each time the query timer expires, the ADVIA Centaur CP system sends a message to the ACCP event log saying the query has timed out and no order record was received for that SID. ACCP sends the next query request.

A remote system that queries for a large quantity of results from the ADVIA Centaur CP system does not require a long timeout if **Q** values are processed. The ADVIA Centaur CP system sends the first result to a query within 5 minutes using Patient ID, Sample ID, Test Name, and Results Date with a system that has nearly full demographic and result databases.

The following figure shows an order record missing a **Q** in the Report Type field. In the example, the response from the remote system contains an **O** instead of **OIQ** value in the Report Type field in the order record. If the response from the remote system requires more than 30 seconds to complete and the query timeout is set to 30 seconds, the query timer expires and the ADVIA Centaur CP system sends a message to the ACCP event log saying the query has timed out and no order record was received for that SID. The ACCP sends the next query request.



Solution

Replace **O** with **OIQ** in Report Type field of order record and in Result Status field of result record.

Increase the value for the Query Timeout at the LIS tab of the Setup screen.

Invalid Test Order

The ADVIA Centaur CP system receives a valid order for a test. The accepted worklist entry does not display at the Result tab.

Possible Cause: Disabled Test

In Setup, assay tab, details the test has been disabled.

Solution

Enter Master Curve data for the required test.

Do not request disabled tests.

Reject

Possible Cause: SID

The SID of a patient sample matches the SID of a control. Control requests do not get queried for by the ADVIA Centaur CP.

Solution

The ADVIA Centaur CP does not query tests requests for Quality Control samples. You must request Quality Control samples manually at the workstation. Ensure that patient SID values do not match Control sample SIDs.

Rejected Test Order for Patient Sample

The ADVIA Centaur CP system rejects an order for a test on a patient sample.

Possible Causes and Solutions

This section identifies the possible causes and solutions for rejected test orders for patient samples.

<i>Possible Cause</i>	<i>Solution</i>
The order record does not contain the Sample ID (SID). The SID is required.	Enter the SID for the patient sample in the order record.
The patient SID entered is the same as the SID defined for a control or a calibrator.	Check the SID for the patient sample. Edit the SID in the worklist so that it is not the same as the SID for the control or the calibrator.

Test Results Transmitted with Wrong Test Name

The ADVIA Centaur CP system sent the incorrect test name for the selected test results for a particular test.

Possible Cause: LIS Code

The LIS code of the ADVIA Centaur CP does not match the LIS code expected by the LIS.

Solution

Check the LIS Code in the assay setup tab. Enter the correct LIS code.

Message Received Caused Fatal Parse Error

The message received by the ADVIA Centaur CP system caused a fatal parse error. The event logged is for an unexpected sequence number, but the sequence number for the record is correct.

Possible Cause: Carriage Return

An application layer record does not terminate with a carriage return (<CR>). If a <CR> is absent, the next record appends to the previous record and the system misinterprets the rest of the message structure.

Solution

Terminate each application layer record with a <CR>.

Possible Cause: Incorrect Sequence Number

An incorrect sequence number is sent with the application layer record.

Solution

Contact your LIS Administrator to check the LIS format.

Invalid ASTM Record Type Received

Only some of the information in a patient record received by the ADVIA Centaur CP system displays in the worklist, and the Event Log shows the system received an invalid ASTM record type.

Possible Cause: Inactive Tests

Not all of the tests in the patient order record are active.

Solution

Verify the status of the tests in the patient order record at the Result tab. Enable the test if necessary.

Possible Cause: Transmission Interference

The ADVIA Centaur CP system received invalid records due to transmission interference.

Solution

The cable between the ADVIA Centaur CP system and the remote system is damaged. Check the continuity of the wires. Replace the cable if necessary.

Possible Cause: Carriage Return

A carriage return (<CR>) is embedded in the demographic data in the fields of the patient record. This truncates the application layer record, and the ADVIA Centaur CP system interprets the remainder as another record. The application layer level uses a <CR> to terminate an application layer record. If required elsewhere in any record, use ASTM escape sequences.

Solution

Remove the <CR> from the demographic data fields of the patient record. Use ASTM escape sequences for any required carriage returns.

Unexpected Data Received in Response to a Query

The remote system received unexpected data from the ADVIA Centaur CP system in response to a query.

Possible Cause: Query Record

The query record is invalid. The following figures show examples of invalid and valid query records.

The following figure shows a query record with a missing component delimiter. A component delimiter should precede the Sample ID in the Starting Range ID Number field. In this example, the Sample ID is interpreted as a Patient ID.

```
Q1|SID10768|ALL| || || ||O<CR>
```

The following figure shows the same query record with the missing component delimiter added.

```
Q1|^SID10768|ALL| || || ||O<CR>
```

The following figure shows a list and a range for the same field. Do not mix lists and ranges. The remote system interprets this query as one closed range of PID 1000000 to PID 199999 that includes REQ1241 and two open-ended ranges that begin with REQ4464 and REQ4465.

```
Q1|PID100000^REQ1241^REQ4464^REQ4465|PID199999|ALL| || || ||O<CR>
```

The following figure shows the same query record with a valid list.

```
Q1|^REQ1241^REQ4464^REQ1241^REQ4465|ALL| || || ||O<CR>
```

The following figure shows the same query record with a valid range.

```
Q1|PID100000|PID199999|ALL| || || ||O<CR>
```

The following figure shows a query record with an unsupported Request Information Status Code field. The ADVIA Centaur CP system does not support the ASTM option of returning only demographic data. The ADVIA Centaur CP system replies to the query shown in the example with a header record and a termination record with a value of **Q** in the Termination Code field.

```
Q1|PID1123|ALL| || || ||D<CR>
```

The following figure shows the same query record with a supported Request Information Status Code field.

```
Q1|PID1123|ALL| || || ||O<CR>
```

Solution

Review the protocol for query records and revise the record. Refer to Section 3, *ASTM*, for more information about the query record.

Delay Between Messages Sent by the ADVIA Centaur CP System

There is a long time between messages sent by the ADVIA Centaur CP system.

Possible Causes and Solutions

This section identifies the possible causes and solutions for delays between messages sent by the ADVIA Centaur CP system. To analyze the problem and find the possible cause, check the communication status and event screens for data link, incoming, and outgoing messages.

<i>Possible Cause</i>	<i>Solution</i>
The remote system does not terminate its query responses with an F , Q , or I code in the Termination Code field of the termination record.	Always terminate queries with the appropriate termination code.
The connecting port on the remote system is not enabled.	Enable the connecting port on the remote system.
The remote system is not terminating its last session at the data link layer.	Always use a terminate session message to relinquish line control when a session is complete.
The cable between the ADVIA Centaur CP system and the remote system is not correctly connected.	Correctly connect the cable. Refer to Section 3, <i>ASTM</i> , for more information about the physical layer.
The cable between the ADVIA Centaur CP system and the remote system is damaged.	Refer to Section 3, <i>ASTM</i> , for more information about the electrical characteristics. Check the continuity of the wires with a breakout box. Replace the cable if necessary.

Automatic Query is Malfunctioning

The automatic query incremental worklist request option is malfunctioning.

Possible Causes and Solutions

This section identifies the possible causes and solutions for incremental worklist malfunctioning.

<i>Possible Cause</i>	<i>Solution</i>
The System Automatically Queries Host for Worklist option is not enabled.	Select System Automatically Queries Host for Worklist at the LIS tab of the Setup screen.
The remote system is not using Specimen IDs in its responses to automatic queries.	The remote system must use valid Specimen IDs in its order records.

Automatic Result Reporting is Malfunctioning

The automatic result reporting is malfunctioning.

Possible Causes and Solutions

This section identifies the possible causes and solutions for automatic result reporting malfunctioning.

<i>Possible Cause</i>	<i>Solution</i>
The System Automatically Sends All Results except Results on Hold option for result reporting is not enabled.	Select System Automatically Sends All Results except Results on Hold at the LIS tab of the Setup screen.
The system is holding test results for operator review and release.	Release the results.

No Communications at All

The ADVIA Centaur CP system and the LIS are not communicating.

Possible Causes and Solutions

This section identifies the possible causes and solutions for no communications at all.

Possible Cause	Solution
The cable is connected to the wrong port.	Connect the cable to the LIS port.
The cable between the ADVIA Centaur CP system and the remote system is not correctly connected.	Correctly connect the cable. Refer to Section 3, <i>ASTM</i> , for more information about the physical layer.
The ADVIA Centaur CP system and the remote system communication parameters do not match.	Ensure that the settings at the Setup – LIS tab match the remote system configuration.
The cable between the ADVIA Centaur CP system and the remote system is damaged.	Refer to Section 3, <i>ASTM</i> , for more information about the electrical characteristics. Check the continuity of the wires with a breakout box. Replace the cable if required.
The LIS port is malfunctioning.	Call for technical assistance. Refer to Appendix B, <i>Service</i> .
The connecting port on the remote system is not enabled, The Connection is not turned on in the Setup LIS screen.	Enable the connecting port on the remote system. Turn on the LIS connection in Setting, LIS
The ADVIA Centaur CP system and the remote system communication parameters do not match.	Check for multiple character errors. Ensure that the settings at LIS tab of the Setup screen match the remote system configuration.

Appendix A: Safety Information

Protecting Yourself from Biohazards

This information summarizes the established guidelines for handling laboratory biohazards. This summary is based on the guidelines developed by the Centers for Disease Control, the Clinical and Laboratory Standards Institute Document M29-A3, *Protection of Laboratory Workers from Occupationally Acquired Infections*, and the Occupational Safety and Health Administration's Bloodborne Pathogens Standard.¹⁻³

Use this summary for general information only. It is not intended to replace or supplement your laboratory or hospital biohazard control procedures.

By definition, a biohazardous condition is a situation involving infectious agents biological in nature, such as the hepatitis B virus, the human immunodeficiency virus (HIV), and the tuberculosis bacterium. These infectious agents may be present in human blood and blood products and in other body fluids.

The following are the major sources of contamination when handling potentially infectious agents:

- needlesticks
- hand-to-mouth contact
- hand-to-eye contact
- direct contact with superficial cuts, open wounds, and other skin conditions that may permit absorption into subcutaneous skin layers
- splashes or aerosol contact with skin and eyes

To prevent accidental contamination in a clinical laboratory, strictly adhere to the following procedures:

- Wear gloves while servicing parts of the instrument that have contact with body fluids such as serum, plasma, urine, or whole blood.
- Wash your hands before going from a contaminated area to a non-contaminated area, or when you remove or change gloves.
- Perform procedures carefully to minimize aerosol formation.
- Wear facial protection when splatter or aerosol formation are possible.
- Wear personal protective equipment such as safety glasses, gloves, lab coats or aprons when working with possible biohazard contaminants.
- Keep your hands away from your face.
- Cover all superficial cuts and wounds before starting any work.

- Dispose of contaminated materials according to your laboratory's biohazard control procedures.
- Keep your work area disinfected.
- Disinfect tools and other items that have been near any part of the instrument sample path or waste area with 10% v/v bleach.
- Do not eat, drink, smoke, or apply cosmetics or contact lenses while in the laboratory.
- Do not mouth pipet any liquid, including water.
- Do not place tools or any other items in your mouth.
- Do not use the biohazard sink for personal cleaning such as rinsing coffee cups or washing hands.

To prevent needlestick injuries, needles should not be recapped, purposely bent, cut, broken, removed from disposable syringes, or otherwise manipulated by hand.

References

1. Centers for Disease Control. 1988. Update: Universal precautions for prevention of transmission of human immunodeficiency virus, hepatitis B virus and other bloodborne pathogens in healthcare settings. MMWR, 37:377–382, 387, 388.
2. National Committee for Clinical Laboratory Standards. Protection of laboratory workers from instrument biohazards and infectious disease transmitted by blood, body fluids, and tissue; approved guideline. NCCLS Document M29-A. Villanova (PA): NCCLS; 1997 Dec. 90p.
3. Federal Occupational Safety and Health Administration. Bloodborne Pathogens Standard. 29 CFR 1910. 1030.

Appendix B: Support Information

This section provides the following information:

- address of the Bayer authorized representative, which is the Bayer contact within the European community
- addresses for obtaining service and technical information

Bayer Authorized Representative

Bayer Diagnostics Europe Limited
Chapel Lane, Swords, Co. Dublin, Ireland

Addresses

For technical assistance contact your local technical support provider. For customer service or additional information contact your local technical support distributor.

Bayer S.A.
División Diagnósticos
Ricardo Gutiérrez 3652
B1605EHD Munro - Buenos Aires
Argentina
+54 (0) 11-4762-7000

Bayer Australia Limited
ABN 22 000 138 714
Diagnostics Business Group
2 Keith Campbell Court
Scoresby Victoria 3179
Australia
+61 (0) 3-9212-8444

Bayer Austria GesmbH
Geschäftsbereich Diagnostika
Lerchenfelder Gürtel 9-11
A-1164 Wien, Austria
+43 (0) 1-711-46-2424

Bayer s.a.-n.v.
Division Diagnostics
Avenue Louise 143 Louizalaan
1050 Bruxelles-Brussel,
Belgium
+32 (0) 2-535-66-81

Bayer S.A.
Produtos Diagnósticos
Rua Domingos Jorge 1100
04779-900 - São Paulo - SP
Brazil
+55-11-5694-5166

Bayer Inc.
77 Belfield Road
Toronto, Ontario
Canada M9W 1G6
416-248-0771

Bayer s.r.o.
Oddílení Diagnostika
Zastoupení pro Ěskou
a Slovenskou republiku
Litvínovská 609/3
CZ-190 21 Praha 9
Ěská republika
+420 266101442

Bayer S.A.
División Diagnóstica
A.A. 80387
Av. De las Américas No 57-52
Santafé de Bogotá, D.C. – Colombia
+571 (09) 423-4199

Bayer HealthCare - Diagnostics
Nørgaardsvej 32
DK-2800 Lyngby, Denmark
+45-45-23-50-00

Bayer OY
Suomalaistentie 7
FIN 02270 Espoo, Finland
+358-9-887-887

Bayer Diagnostics
49-51 Quai de Dion Bouton
92815 Puteaux Cedex, France
+33 (0) 1-49-06-56-00

Bayer Vital GmbH
Diagnostika
Siemensstraße 3
D-35463 Fernwald, Germany
+49 (0) 641-4003-0

Bayer Hellas AG
18-22 Sorou str.
GR-151 25 Marousi
Athens, Greece
+30 (0) 210 6187 500

Bayer HealthCare Co. Ltd.
20/F Gee Chang Hong Centre
65 Wong Chuk Hang Road
Hong Kong
852-28147337

Bayer Hungária Kft.
Alkotás Point
H-1123 Budapest, Hungary
Alkotás u. 50
+36 (06) 1 4874 100

Bayer Diagnostics India Limited
589, Sayajipura
Ajwa Road
Baroda – 390 019
Gujarat, India
+91 (0) 26-5562720

Bayer S.p.A.
Divisione Diagnostici
Via Grosio 10/4
20151 Milano, Italia
+39-2-3978-1

Bayer Medical Ltd.
1-19-15, Ebisu
Shibuya-Ku
Tokyo 150-0013, Japan
+81 (0) 3-440-2411

Bayer Diagnostics Korea Limited
Kye Myung Bldg. 4F
Myungil-Don, 48-7
Kangdong-Ku
Seoul 134070
Korea
+82 (0) 2-428-5987

Bayer (Malaysia) Sdn. Bhd
11th Floor
Wisma Damansara
Jalan Semantan
50490 Kuala Lumpur
Malaysia
+603 (0) 62093088

Bayer de México, S.A. de C.V.
División Diagnósticos
Avenida Coyocacán No. 1553
Colonia del Valle
Código Postal 03100
Delegación Benito Juárez
México, D.F.
Centro de Atención Telefónica (CAT)
+52 55 57 28 33 12
+52 55 57 28 33 06

Bayer B.V.
Health Care
Division Diagnostics
Energieweg 1
3641 RT Mijdrecht
The Netherlands
+31 (0) 297-280666

Bayer New Zealand Ltd
Diagnostics Business Group
3 Argus Place,
Glenfield,
Auckland, New Zealand
+64 9-441-8549

Bayer AS
Drammensvn. 147 B
NO-0277 Oslo, Norway
+47 24 11 18 00

Bayer Sp. Z o.o.
Al. Jerozolimskie 158
02-326 Warszawa, Polska
+48 (0) 22-572-3500

Bayer Diagnostics Europe Ltd.
(sucursal em Portugal)
Rua da Quinta do Pinheiro, 5
2795-653 Carnaxide, Portugal
+351-21-416-4227

Bayer Puerto Rico Inc.
Diagnostics Division
Victoria Industrial Park
Building 1
Carolina, Puerto Rico
787-752-8989

Bayer (South East Asia) Pte Ltd.
No. 9 Benoi Sector
Singapore 629844
+65-6261-3389

Bayer (Pty) Ltd.
Healthcare Division
27 Wrench Road
Isando 1600, South Africa
+27 (0) 11-921-5048

Química Farmacéutica Bayer, S.A.
División Diagnósticos
c/ Pau Claris, 196
08037 Barcelona, España
+34-93 4956500

Bayer AB
Drakegatan 1
S-402 24 Göteborg, Sweden
+46-31-83-98-00

Bayer (Schweiz) AG
GB DS
Grubenstrasse 6
CH-8045 Zürich
+41 (0) 1-465-81-11

Bayer plc
Diagnostics Division
Bayer House
Strawberry Hill
Newbury, RG14 1JA
United Kingdom
+44 (0) 1635-563000

Bayer HealthCare LCC
Subsidiary of Bayer Corporation
511 Benedict Avenue
Tarrytown, NY 10591-5097 USA
914-631-8000

Appendix C: Messages

This appendix shows examples of ADVIA Centaur CP system and LIS messages from the data link and application layers.

Characters from the Application Layer

Automatic Worklist Request and Response for a Patient Sample

The following example shows an automatic worklist request and response for a patient sample. In this example, the **OIQ** value in the order record of the LIS reply, indicates a worklist entry that is transmitted to the ADVIA Centaur CP system in response to a query. The absence of **F** in the termination record indicates that this is not the last item in response to the query and no additional queries are sent.

ADVIA Centaur CP System	LIS
H \^& ACCP1 Host P 1<CR>	→
Q 1 ^P0650 ALL O<CR>	→
L 1<CR>	→
←	H \^&<CR>
←	P 1 20 Jones^Alan^B 19560519 M Jones^Alan^B General_Ward<CR>
←	O 1 P0650 ^T4^^TU^^fti R O\Q <CR>
←	L 1<CR>

Automatic Worklist Request and Response with No Demographic or Pending Test Information

The following example shows an automatic worklist request and response when the remote system has no demographic or pending test information. The request specifies sample ID SID10768. In this example, the presence of **I** in the termination record from the LIS indicates that no information is available to send in response to the query.

ADVIA Centaur CP System	LIS
H \^& ACCP1 Host P 1<CR>	
Q 1 ^SID10768 ALL O<CR>	
L 1<CR>	
	H \^&<CR>
	L 1 I<CR>

Manual Worklist Request and Response with No Demographic or Pending Test Information

The following example shows a manual worklist request and response when the remote system has no demographic or pending test information. The request specifies sample ID SID10768. In this example, the presence of **I** in the termination record from the LIS indicates that no information is available to send in response to the query.

ADVIA Centaur CP System	LIS
H \^& ACCP1 Host P 1<CR>	
Q 1 ALL ALL O<CR>	
L 1 N<CR>	
	H \^&<CR>
	L 1 I<CR>

Manually Initiated Worklist Query With Demographic and Assay Response

ADVIA Centaur CP System

LIS

```

H|\^&|||ACCP1||||Host||P|1<CR>
Q|1|ALL||ALL|||||O<CR>
L|1<CR>
H|\^&|||Host||||ACCP1<CR>
P|1|PID1123|||Peters^Cassie||19721221|F||||DR1265A<CR>
O|1|REQ1241||^T3^T4^TSH|R|||||||O\Q<CR>
O|2|REQ1241||^HCG|R|||||||O\Q<CR>
P|2|PID4423|||Jacobs^Hal^M||19660822|M||||DR4267J|||||ER1<CR><CR>
O|1|REQ4464||^CEA|R|||||||O\Q<CR>
O|2|REQ4465||^PSA|R|||||||O\Q<CR>
L|1|F<CR>

```

LIS Cancellation of a Scheduled Test

The following example shows a cancellation by an LIS of a scheduled test for a specific sample.

ADVIA Centaur CP System

LIS

```

H|\^&<CR>
P|1<CR>
O|1|REQ1241||^ThCG|R||||C|||||||O<CR>
L|1<CR>

```

Automatic Result Reporting

The following example shows automatic result reporting.

ADVIA Centaur CP System

LIS

H|\^&|||ACCP1||||Host|P|1<CR>—————→

P|1|PID4423|||Jacobs^Hal^M||19660822|M||||DR4267J||||| ||||WARD 3<CR>—————→

C|1||Repeat draw.|G<CR>—————→

O|1|REQ4464||^C^CEA|R|||||||||||||F<CR>—————→

R|1|^C^CEA^^1^DOSE|6.62|ng/mL||H||F|||199209270807<CR>—————→

R|2|^C^CEA^^1^RLU|36632||||F|||199209270807<CR>—————→

C|1|I|Repeated|I<CR>—————→

C|2|I|Above Check|I<CR>—————→

L|1<CR>—————→

Operator-Initiated Transmission of Results

The following example shows an operator-initiated transmission of results from the ADVIA Centaur CP system.

ADVIA Centaur CP System

LIS

```

H|\^&|||ACCP1|||||Host|P|1<CR>—————→
P|1|PID1123|||Peters^Cassie||19721221|F||||DR1265A<CR>—————→
O|1||REQ124|||T4^T3^TUP^ThCG|R|||||||F<CR>—————→
R|1|^T4^1^DOSE|5.8|ug/dL||||F||||199209270807<CR>—————→
R|2|^T4^1^RLU|54688||||F||||199209270807<CR>—————→
R|3|^T3^1^DOSE|1.27|ng/mL||||F||||199209270807<CR>—————→
R|4|^T3^1^RLU|14527||||F||||199209270807<CR>—————→
R|5|^TUP^1^DOSE|11.626|uIU/mL||||F||||199209270808<CR>—————→
R|6|^TU^1^RLU|337704||||F||||199209270808<CR>—————→
R|7|^ThCG^1^DOSE|4.2|mIU/mL||||F||||199709270808<CR>—————→
R|8|^ThCG^1^RLU|24342||||F||||199709270808<CR>—————→
L|1<CR>—————→

```

LIS Query for Results by Starting Range

The following example shows an LIS query for results with a single value in the Starting Range ID Number field. The ADVIA Centaur CP system transmits all results with a PID greater than or equal to PID1123 to the LIS.

ADVIA Centaur CP System

LIS

```

← H|\&<CR>
← Q|1|PID1123||ALL|||||F<CR>
← L|1<CR>

H|\&||ACCP1||||Host||P|1<CR>
P|1|PID1123||Peters^Cassie|19721221|F||||DR1265A<CR>
O|1|REQ1241||^T4^T3^TUP^ThCG|R|||||||F<CR>
R|1|^T4^1^DOSE|5.8|ug/dL||||F\R||||199709270807<CR>
R|2|^T4^1^RLU|54688||||F\R||||199709270807<CR>
R|3|^T3^1^DOSE|1.27|ng/mL||||F\R||||199709270807<CR>
R|4|^T3^1^RLU|14527||||F\R||||199709270807<CR>
R|5|^TU^1^DOSE|11.626|uIU/mL||||F\R||||199709270808<CR>
R|6|^TUP^1^RLU|337704||||F\R||||199709270808<CR>
R|7|^ThCG^1^DOSE|4.2|mIU/mL||||F\R||||199709270808<CR>
R|8|^ThCG^1^RLU|24342||||F\R||||199709270808<CR>
L|1|F<CR>

```


LIS Query for Results by Open Range and List of Tests

The following example shows an LIS query for results by specifying an open range on the SIDs and a list of tests. Note that the order of tests in the order record and the order of result records returned are not necessarily the same. Note the **FIR** in the Result Status field of the result records, which indicates final results, previously transmitted.

ADVIA Centaur CP System

LIS

← H| $\wedge$ <CR>

← Q1| ^REQ1241|| ^^^T4\ ^^^T3\ ^^^TSH||| | F<CR>

← $L/1\langle CR \rangle$

$$H \vee \& || ACCP1 || || Host | P1 < CR \rangle \longrightarrow$$

P|1PID1123||Peters^Cassie||19721221F||||DR1265A<CR>

O1|REQ1241||^T4\^T3\^TSH|R|||||F<CR>

R|1^^T4^^1^DOSE|5.8ug/dL|||F\R|||199209270807<CR>—————>

R|2|^^T4^^1^RLU|54688|||F\R|||199209270807<GR>

R|3|^^^T3^^^1^DOSE|1.27|ng/mL|||F\R|||199209270807<GR>

R|4|^^T3^^1^RLU|14527|||F\R|||199709270807<CR>

R|5|^^TSH^^1^DOSE|11.626|uIU|||F\R|||199709270808<GR>—————→

R|6|^^TSH^^1^RLU|337704|F|R||199709270808<GR>—————→

$$L|1|F\langle CR\rangle \longrightarrow$$

The following example shows an LIS query for results by a range of dates and times.

LIS

086D0004-01, Rev. A

ADVIA Centaur CP System Response to a Result Query in Error

The following example shows an ADVIA Centaur CP system response to an LIS query in error. Note that the value of the Request Information Status Code field of the query record is an option not supported by the ADVIA Centaur CP system. Note also the value **Q** in the Termination Code field of the termination record.

ADVIA Centaur CP System

LIS

```

←----- H|\^&<CR>
←----- Q|1|ALL||ALL||199709270807|199709270808||||D<CR>
←----- L|1<CR>

H|\^&|||ACCP1||||Host||P|1<CR>-----→
L|1|Q<CR> -----→

```

ADVIA Centaur CP System Response to an LIS Cancellation of Query

The following example shows the ADVIA Centaur CP system response to an LIS cancellation of query. Note that the first query has no restrictions. If the ADVIA Centaur CP result database is nearly full, the LIS could decide that the query timed out before the ADVIA Centaur CP system responds with its first record.

ADVIA Centaur CP System

LIS

```

←----- H|\^&<CR>
←----- Q|1|ALL||ALL||||||F<CR>
←----- L|1<CR>

```

Ten minutes pass. The operator of the LIS tires of waiting for data and cancels the query to make a more selective query. Note the **A** in the Request Information Status Code field in the second query record.

```

←----- H|\^&<CR>
←----- Q|1|||||||A<CR>
←----- L|1<CR>

```

The ADVIA Centaur CP system stops processing the initial query and is ready to receive a new query.

ADVIA Centaur CP System Response to the LIS Cancellation of Query

The following example shows the ADVIA Centaur CP system response when the LIS cancels a query in progress. The first query has the incorrect month for the Beginning Request Results Date field and the operator of the LIS observes that received data has incorrect dates. The operator of the LIS cancels the query with a query cancel. Note that after the query cancel, the remote system may receive some data before the response to the query is canceled. Note also the **R** in the Termination Code field of the termination record from the ADVIA Centaur CP system.

ADVIA Centaur CP System	LIS
	← H \^&<CR>
	← Q 1 ALL ALL 199708270800 199709271200 F<CR>
	← L 1<CR>
H \^& ACCP1 Host P 1<CR>	→
P 1 PID1123 Peters^Cassie 19721221 F DR1265A<CR>	→
O 1 REQ1241 ^T4^T3^TSH R F<CR>	→
R 1 ^T4^1^DOSE 5.8 ug/dL F\R 199708270807<CR>	→
R 2 ^T4^1^RLU 54688 F\R 199708270807<CR>	→
R 3 ^T3^1^DOSE 1.27 ng/mL F\R 199709270807<CR>	→
R 4 ^T3^1^RLU 14527 F\R 199709270807<CR>	→
	← H \^&<CR>
	← Q 1 A<CR>
	← L 1<CR>
R 5 ^TSH^1^DOSE 11.626 uIU/mL F\R 199708270808<CR>	→
R 6 ^TSH^1^RLU 337704 F\R 199708270808<CR>	→
L 1 R<CR>	→

Characters from the Data Link and Application Layers

In this section, transmissions from the ADVIA Centaur CP system are preceded by **A**: and transmissions from the LIS are preceded by **L**:

The following table shows the non-printable ASCII characters that are represented by ASCII mnemonics between angle brackets.

<i>Mnemonic</i>	<i>Decimal Value</i>
<STX>	2
<ETX>	3
<EOT>	4
<ENQ>	5
<ACK>	6
<BEL>	7
<HT>	9
<LF>	10
<VT>	11
<CR>	13
<NAK>	21
<ETB>	23
<SP>	32

Query for Orders from the ADVIA Centaur CP System

The following example shows a query for orders from the ADVIA Centaur CP system and the subsequent order from an LIS.

```

A: <ENQ>
L: <ACK>
A: <STX>1H|\^&|||ACCP1||||Host||P|1<CR><ETX>E5<CR><LF>Q|1|
REQ3051||ALL|||||O<CR><ETX>D7<CR><LF>L|1<CR><ETX>3C<CR><L
F>
L: <ACK>
A: <EOT>
L: <ENQ>
A: <ACK>
L: <STX>1H|\^&|||Host||||ACCP1||P|1<CR>P|1|P1234567015|||Jacobs^
Hal||19480612|M||||MD_DOC|||||||WARD<CR>C|1|^Patient comment15|
G<CR>C|2|^Patient comment15|G<CR>O|1| REQ3051||ALL|||||O
||^TSH\^^LH\^^THCG\^^PRL|||||||||||O\Q<CR>
A: <ACK>
L: <STX>2L|1|F<CR><ETX>04<CR><LF>
A: <ACK>
L: <EOT>

```

Automatic Worklist Request Queries Transmitted from the ADVIA Centaur CP System

There are three types of query modes to select from:

- The Single query, the query is for a single request and may be followed by another single query, by an order record, or any other record.
- The second type of query is All, the query may be for multiple samples in one record and may be followed by any other record.
- The third type of query is ADVIA Centralink. The query is for a single request and must be followed by an order record before another query can be made. If an order record is not received then the event is logged to the ACCP event log and after the query times out a new query record can be sent.

The following example shows the Single query mode

ADVIA Centaur CP System**LIS**

```

H|\^&|||ACCP1||||HOST||P|1<CR> ----->
Q|1|^SID1001||ALL|||||O<CR>----->
L|1|N|<CR> ----->
H|\^&|||ACCP1||||HOST||P|1<CR> ----->
Q|1|^SID1002||ALL|||||O<CR> ----->
L|1|N|<CR> ----->
<----- H|\^&|||HOST||||ACCP1||P|1<CR>
<----- P|1|PatID01|||Conti^Biagio^S||19741001|M||||Martinez|||||||WestWing<CR>
<----- O|1|SID1001|||^^^T4^^R|||||||O<CR>
<----- L|1|N<CR>

H|\^&|||ACCP1||||HOST||P|1<CR> ----->
Q|1|^SID1003||ALL|||||O<CR> ----->
L|1|N|<CR> ----->
H|\^&|||ACCP1||||HOST||P|1<CR> ----->
P|1|PatID02|||Steckel^Mark^A||19741001|M||||Jones|||||||Emergency<CR>----->
O|1|SID1000|||^^^AFP||R||||O/Q<CR>----->
R|1|^AFP^^^DOSE|13.1|IU/ml||H||F|||20050506123145<CR>----->
L|1|N|<CR> ----->
<----- H|\^&|||HOST||||ACCP1||P|1<CR>
<----- P|1|PATD02|||Jones^Mary||19640521|F|<CR>
<----- O|1|SID1002|||^^^T4^^R|||||||O<CR>
<----- L|1|N<CR>

```

The following example shows the All query mode:

ADVIA Centaur CP System

LIS

```

H|\^&|||ACCP1||||HOST||P|1<CR>—————→
Q|1|^SID1001||ALL|||||||O<CR>  —————→
L|1N|<CR> —————→

H|\^&|||ACCP1||||HOST||P|1<CR>—————→
Q|1|^SID1002||ALL|||||||O<CR>—————→
L|1N|<CR>  —————→

←————— H|\^&||HOST||||ACCP1||P|1<CR>
←————— P|1|<CR>

←————— O|1|SID1001||^T4^R|||||||O<CR>
←————— L|1|I<CR>

←————— H|\^&||HOST||||ACCP1||P|1<CR>
←————— P|1|<CR>

←————— O|1|SID1002||^T4^R|||||||O<CR>
←————— L|1|I<CR>

```


The following example shows the ADVIA CentraLink query mode:

ADVIA Centaur CP System

LIS

```

H|\^&|||ACCP1||||HOST||P|1<CR>—————→
Q|1|^SID1001||ALL|||||O<CR>—————→
L|1N|<CR>—————→

←————— H|\^&||HOST||||ACCP1||P|1<CR>
←————— P|1|<CR>

←————— O|1|SID1001||^T4^dilution profile^DilProName|R|||||||O<CR>
←————— L|1|I<CR>

H|\^&|||ACCP1||||HOST||P|1<CR>—————→
Q|1|^SID1002||ALL|||||O<CR>—————→
L|1N|<CR>—————→

←————— H|\^&||HOST||||ACCP1||P|1<CR>
←————— P|1|<CR>

←————— O|1|SID1002||^T4^R|||||||O<CR>
←————— L|1|I<CR>

```

Transmission from the ADVIA Centaur CP System with No Response

The following example shows a transmission from the ADVIA Centaur CP system with no response from the remote system. After the initial enquiry (<ENQ>) timeout, five more enquires must timeout before reaching the retry limit and then an error displays at the event log.

```

A: <ENQ> <EOT> <ENQ> <EOT> <ENQ> <EOT> <ENQ> <EOT>
    <ENQ> <EOT> <ENQ> <EOT>

```

The ADVIA Centaur CP system continues the enquiries until it receives a successful response (<ACK>) from the remote system.

Appendix D: Records

This appendix shows examples of the ADVIA Centaur CP communications protocol records and messages. Refer to Section 3, *ASTM*, for an explanation of each record type.

Header Record

The following example shows the minimum header record that the ADVIA Centaur CP system recognizes.

```
H|\^&|<CR>
```

The following example shows the minimum header record that the ADVIA Centaur CP system transmits.

```
H|\^&|||ACCP1||||Host|P|1<CR>
```

The following example shows a header record from an LIS using Sender and Receiver ID fields.

```
H|\^&|||Host||||ACCP1<CR>
```

Patient Record

The following example shows the minimum patient record that the ADVIA Centaur CP system recognizes. The ADVIA Centaur CP system transmits the minimum record as the patient record for a control sample worklist order or a control sample result.

```
P|1<CR>
```

The following example shows the minimum patient record for a patient sample that the ADVIA Centaur CP system transmits. Note the **U** (Unknown) in the Patient Sex field; the system requires a value in this field.

```
P|1|||||U<CR>
```

The following example shows a patient record with all fields that the ADVIA Centaur CP system uses.

```
P|1|PID456|||Smith^Gretchen^A||19760423|F||||ID32|||||||ER<CR>
```

Patient Comment Record

The ADVIA Centaur CP system always transmits the Comment Text field and the Comment Type field or it does not send the record. The system does not send the Comment Source field. The following example shows a transmitted patient comment record.

```
C|1||^Insufficient sample. Redraw.|G<CR>
```

The following example shows a patient comment record from an LIS with the least number of fields.

```
C|1||^Contact Dr. Conti, x2777|G<CR>
```

The following example shows a patient comment record from an LIS with the maximum number of fields.

```
C|1|L|^Contact x3378|G<CR>
```

Order Record

The following example shows the minimum order record that the ADVIA Centaur CP system transmits. The ADVIA Centaur CP system always transmits the Sample ID, Test Name or LIS Code, Priority, and Report Type fields or it does not send the record.

```
O|3|SID4728||^T4|S|||||||O<CR>
```

The following example shows an order record with all fields that the ADVIA Centaur CP system uses. Note the **Q** (Q/C Test Specimen) in the Action Code field.

```
O|2|P016101||^T3|R|||||Q|||||||O<CR>
```

The following example shows an order record for a patient sample worklist order, sent by an LIS to cancel the FSH test for the sample with a sample ID of 3A6BZ201. Note the **C** (Cancel Request) in the Action Code field.

```
O|1|3A6BZ201||^FSH|R|||||C|||||||O<CR>
```

The following example shows an order record for the results of tests performed on a stat patient sample, sent by the ADVIA Centaur CP system in response to a query. Note the **S** (Stat) in the Priority field. Result records for each test and zero or more result comment records for each result record follow.

```
O|1|SID7201||^LH^HCG^TSH^PROLACT|S|||||||F\Q<CR>
```

The following example shows the ADVIA Centaur CP system transmitting an order record to inform a remote system that it is rejecting the worklist order from the remote system and it is not processing the worklist order. Note the **X** (Work Cannot Be Done) in the Report Type field.

```
O|1|SID7201^0056^E||^LH^HCG^TSH^PROLACT|R|||||||X<CR>
```

Result Record

The ADVIA Centaur CP system rejects all result records.

The following example shows the minimum result record that the ADVIA Centaur CP system transmits. The system always sends an LIS code, result aspects, a measurement value, a result status, and a date test completed. This result is sent in response to a query.

```
R|1|^^^TU^^^INDX|0.822||||F\Q|||199708232138<CR>
```

The following example shows a result record with all fields that the ADVIA Centaur CP system uses. In addition to the fields always sent, the Units and Result Abnormal Flags fields have values. The result is low. Note the **L** (Below Reference Range) in the Result Abnormal Flags field.

```
R|1|^^^T4^^^1^DOSE|4.2|ug/dL||L||F|||199709060902<CR>
```

The following example shows a result record in which the result is a value beyond the range of the Master Curve, a value in the Result Abnormal Flags field, and an empty Units field. This result is greater than the maximum range of the Master Curve. It was previously transmitted; note the **R** (Results Previously Sent) in the Result Status field.

```
R|1|^^^T4^^^1^DOSE|>21.00|ug/mL||>||F\R|||199709060902<CR>
```

Result Comment Record

ADVIA Centaur CP system transmits additional result abnormal flags in the Comment Text field. The system always transmits the Comment Source, Comment Text, and Comment Type fields. The following example shows a result comment record that the ADVIA Centaur CP system transmits. Note the **I** in the Comment Type field when the system reports a result flag. The ADVIA Centaur CP system ignores all result comment records.

```
C|1|I|Repeated|I<CR>
```

The following example shows a operator-entered result comment record that the ADVIA Centaur CP system transmits. Note the **G** in the Comment Type field when the system reports a user-entered comment.

```
C|1|I|^Gross hemolysis may affect result|G<CR>
```

Query Record

The following example shows a query record initiated from the ADVIA Centaur CP system by an operator. This record requests all worklist information from a remote system and always has the same field values. Note the **O** (Only test Orders and Demographics) in the Request Information Status Code field.

```
Q|1|ALL||ALL||||||O<CR>
```

The following example shows a query record initiated automatically from the ADVIA Centaur CP system after it reads sample barcodes. The ADVIA Centaur CP system sends a query for each sample identified by the system, whether or not an entry exists in the database. These queries always request worklist orders for all tests for a sample.

```
Q1|^SID0782|^SID0782|ALL|||||O<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results with several restrictions. The search returns all results for a sample with a sample ID (SID) of S1234 and all results for a sample with a SID of S4321 completed within the date range from 11:00 p.m. on 31 August 1997, through the hour of 4:00 a.m. on 01 September 1997.

```
Q1|^S1234^S4321|^S1234^S4321|ALL||1997083123|1997090104||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results with several restrictions. The search returns all results for samples within the SID range of S1234 through S1321 and all results for a samples within the SID range of S4321 through S4510. The results for all samples must be completed after 8:00 p.m. on 31 August 1997, and before 6:00 a.m. on 01 September 1997.

```
Q1|^S1234^S4321|^S1321^S4510|ALL||1997083120|1997090106||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results with several restrictions. Note the **F** (Final Results) in the Request Information Status Code field. The case-sensitive search returns all t4 final result replicates within the PID range of P1234 through R and the SID range of SID0600 through SID0700, and resulted on or after the month of June, 1997.

```
Q1|P1234^SID0600|R^SID0700|^^^t4||1997072308\199706||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results for a specific shift. The search returns all results for all samples completed within the date and time range from 8:00 a.m. on 01 October 1997, through the hour of 4:00 p.m. on 01 October 1997.

```
Q1|ALL||ALL||1997100108|1997100116||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results for a specific day for archiving by the LIS. The search returns all results for all samples completed for the day of 14 September 1997.

```
Q1|ALL||ALL||19970914|19970914||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results for specific tests for a month for quality control (QC) analysis. The search returns all results for all T4 and FT4 final results completed for the month of September, 1992.

```
Q1|ALL||^T4^FT4||199709|199709||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results for a range of SIDs. The search returns all results for all tests performed on SIDs from REQ1241 through REQ1300.

```
Q1|^REQ1241|^REQ1300|ALL|||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results for a single PID. The search returns all results for all tests performed on samples associated with the PID 275-47-0149.

```
Q1|275-47-0149|275-47-0149|ALL|||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database for results for a specific test. The search returns all TSH final results in the database.

```
Q1|ALL|^TSH|||||F<CR>
```

The following example shows a query record accepted by the ADVIA Centaur CP system that searches the database to determine which ordered tests are still pending. Note the **I** (In Instrument Pending) in the Request Information Status Code field.

```
Q1|ALL|ALL|||||I<CR>
```

Termination Record

The following example shows the minimum termination record that the ADVIA Centaur CP system recognizes. It is parsed and logged as a normal termination. If the ADVIA Centaur CP system sends a termination record in a message that is in response to a query, the system has not completed its response.

```
L|1<CR>
```

The following example shows a termination record that completes the response to a query. When the ADVIA Centaur CP system sends this record, the system considers its response to be complete, and it is ready to accept a new query.

When the ADVIA Centaur CP system receives a termination record with a termination code of **F**, the system closes any outstanding query it sent and logs the query as complete. If the ADVIA Centaur CP system has sent a query and never receives a termination record with a Termination Code of **F** (Final Record), **Q** (Query in Error), or **I** (No Information Available), the query remains pending until it times out. The ADVIA Centaur CP system does not send any subsequent queries until the pending query times out.

```
L|1|F<CR>
```

The following example shows a termination record that completes a query response that has no data. When sent by the ADVIA Centaur CP system, the ADVIA Centaur CP system has no data to send and considers its response to be complete. It is ready to accept a new query. When received by the ADVIA Centaur CP system, the ADVIA Centaur CP system closes any outstanding query it sent and logs the query as receiving no data.

L|1|I<CR>

The following example shows a termination record that completes a response to a query in error. This is sent by the ADVIA Centaur CP system under two conditions:

- the system receives a second query (other than a query cancellation) before it completes a response to a preceding query.
- the system receives a query with a value in the Request Information Status Code field other than, **F** (Final), **R** (Results Previously Transmitted), **C** (Edited Results), **I** (In Instrument Pending), or **A** (Cancel Transmitted Data).

The ADVIA Centaur CP system treats any queries received as closed and is ready to accept a new query. If received by the ADVIA Centaur CP system, the ADVIA Centaur CP system closes any outstanding query it sent and logs the query as in error.

L|1|Q<CR>

Appendix E: Invalid Characters

This section lists the invalid data characters in ASTM messages at the data link and the application layers. If any invalid characters are present in an ASTM message at the data link or the application layer, the message is not accepted. For more information, refer to *Data Link Messages* in Section 2 of the *ADVIA Centaur CP LIS Interface Guide*.

An X in the data segment or the application layer indicates an invalid character. Characters not listed in this appendix are valid.

Hexadecimal Character	Decimal Character	ASCII Character	Data Segment	Application Layer
00	0	NUL		X
01	1	SOH	X	X
02	2	STX	X	X
03	3	ETX	X	X
04	4	EOT	X	X
05	5	ENQ	X	X
06	6	ACK	X	X
08	8	BS		X
0A	10	LF	X	X
0E	14	SO		X
0F	15	SI		X
10	16	DLE	X	X
11	17	DC1	X	X
12	18	DC2	X	X
13	19	DC3	X	X
14	20	DC4	X	X
15	21	NAK	X	X
16	22	SYN	X	X
17	23	ETB	X	X
18	24	CAN		X
19	25	EM		X
1A	26	SUB		X
1B	27	ESC		X
1C	28	FS		X

<i>Hexadecimal Character</i>	<i>Decimal Character</i>	<i>ASCII Character</i>	<i>Data Segment</i>	<i>Application Layer</i>
1D	29	GS		X
1E	30	RS		X
1F	31	US		X
7F	127	DEL		X
FF	255			X

Appendix F: Glossary

algorithm	A set of rules or processes used for solving a specific problem.
ANSI	American National Standards Institute
application data segment	Subset of an application message that gets passed to or from the data link layer.
application layer	Layer 3 of the ASTM communication protocol. Provides instrument and LIS information services.
application message	Contains data concerned with the system application, such as worklist data or result data.
ASCII	American Standard Code for Information Interchange. A code for representing a set of alphanumeric characters used by information processing systems and communications systems.
ASTM	American Society for Testing and Materials
ASTM E1381-95	Standard document for the physical and data link layers of the ASTM communication protocol, which is titled <i>Specification for Low-Level Protocol to Transfer Messages Between Clinical Laboratory Instruments and Computer Systems</i> .
ASTM E1394-91	Standard document for the application layer of the ASTM communication protocol, which is titled <i>Standard Specification for Transferring Information Between Clinical Instruments and Computer Systems</i> .
automatic result reporting	A system option. When enabled, final results are transmitted from the system as soon as they are added to the results database, without any intervention by the operator.
automatic transfers	Automatic result reporting and automatic worklist entry requests.
automatic worklist entry requests	A system option. When enabled, a query for test orders is transmitted for each barcoded sample scanned and added to the worklist database.
Backus-Naur form	Notation used to define the syntax of a software language.
baud rate	Speed at which data is sent or received when devices are communicating through a serial channel.
BNF	See Backus-Naur form.

buffer	Data storage used to compensate for a difference in the flow rate of information or the time that events occur when transmitting data from one device to another.
cancel query	Stop the response to a query before it is completed. This can be done by the system that sent the query and by the system that received the query.
cancel test	Unschedule a test for a sample. If a test is not running, it is not run. If a test started running, but is not finished, no result is read. If a test is finished, a cancellation has no effect.
CCITT	International Telegraph and Telephone Consultative Committee.
checksum	A variable, two-digit, hexadecimal number included in data link message frames to support error detection.
communications parameters	Characteristics of the communications setup that are configured at the system.
communications port	The RS-232 port that is used to connect a remote device to the system.
concatenate	To link or join two or more character strings into a single character string.
configuration	The settings of software and hardware characteristics of a device.
control character	A character used to provide information about transmitted data or to control data transmission.
data bits	Physical layer parameter used to specify the number of bits that encode a single data character. Can be 7 or 8.
data link layer	Layer 2 of the ASTM communication protocol. Ensures reliable transfer of data across the physical medium.
data link message	Logical unit of data that contains application data segments along with additional data that provides transmission synchronization and error control information.
data segment	Section of a data link frame that contains application message data.
decimal digits	The base 10 number system.
define	To establish a value for a variable or symbol or to establish what the variable represents.
delimiter	The character that marks the beginning or end of a group of data.

demographics	Information about a patient. On the system, this is the patient name, patient ID, birthdate, sex, location, physician, and comments.
deny session message	Data link layer message used by the receiver during the link establishment phase when the receiver cannot immediately receive information.
device	An instrument or computer used by the communications package.
download	To send data from a remote system to the system.
duplexity	A characteristic of a communications link that relates to the direction and timing of data transmission across the physical medium.
enable	A command or condition that permits some specific event to proceed.
flow control	Any method used to regulate incoming data in an effort to have the time and resources to process the incoming data without losing any of it.
frame	Section of a data link message. Consists of a frame header, a data segment, and a frame trailer.
frame sequence number	A number included in a frame to synchronize the proper ordering of frames.
framing error	The start or stop bits of a byte are encountered unexpectedly or are not encountered when expected by the physical layer. The condition is reported to the datalink layer as a framing error. This error is unrelated to a frame sequence number error.
grant session message	Data link layer message used by the receiver during the link establishment phase when the receiver can receive information.
half-duplex	ANSI terminology used to describe a data link duplexity where both stations may transmit, but only one at a time. Same as CCITT simplex.
handshaking	The mechanism for coordinating the transmission of data across a channel.
hexadecimal digits	The base 16 number system.
incremental transfer	The process of sending a single worklist entry or result value from one device to another.
interface	A shared boundary, such as an RS-232 port, that enables the system and other devices to interact.
interframe timeout	The period of time a receiving system waits for the next frame before assuming that the sending system is disabled.

intermessage timeout	Same as interframe timeout.
link establishment phase	First phase of an ASTM data link session. Establishes which system has control of the communications link to transmit data.
link release phase	Third and final phase of an ASTM data link session. Control of the communications link is released during this phase.
LIS	Laboratory information system. A computer used in a clinical laboratory or hospital to store and transmit worklist requests, test results, and patient information to and from laboratory instruments.
local query message	A query transmitted by the system to a remote system. A data string with a specific meaning transmitted over a communications channel. Should always have a descriptive word preceding it, for example, application message.
message transfer phase	Second phase of an ASTM data link session. Application message data is transmitted during this phase.
modulo	The number of values a variable has before it returns to a value of zero and repeats the set of values. For example, a single decimal digit has ten values, from zero to nine.
negative acknowledgement message	Response sent by the data link layer of the receiving system to indicate that a frame was invalid.
null field	A storage area in a data structure, such as a record or database table, that is empty.
null string	An empty character string.
parity checking	A character error detection code that uses the binary digits of the data bits and any existing parity bit, in which the total number of 1s in each encoded character is even or odd. The parity bit is set to 1 or 0, or the parity bit is non-existent. The names of the three parity checking schemes are EVEN, ODD, or NONE, respectively.
parse	To associate character strings (data) with the component names of the specific message.
physical layer	Layer 1 of the ASTM communication protocol. Concerned with the electrical, mechanical, and timing aspects of signal transmission over a medium.
PID	Patient ID. Any character string used by the operators of the system to identify a patient.

point-to-point	A configuration in which exactly two stations share a transmission path.
positive acknowledgement message	Response sent by the data link layer of the receiving system to indicate that a frame was valid.
positive acknowledgement with interrupt message	Response sent by the data link layer of the receiving system to indicate that a final frame was valid and the receiving system has a data link message to transmit.
profile	A defined group of tests that you can select for a sample.
protocol	The set of conventions that applies to the format and relative timing of the way messages transmit between the communicating devices.
query	A request for information from the receiving system. This may refer to the record that defines the request or to the message that contains the query record.
receiver	The computer system that receives data segments during a data link session.
remote query	A query sent by a remote system and received by a system.
remote system	The device on the other end of the physical connection between two communicating devices. Generally in this document, the remote system is an LIS, a general purpose computer, or test equipment that is connected to the system.
replicate	A single instance of data or related set of data with a one-to-one relationship. In the context of a database, a single row of data in a table.
request session message	A data link layer message used by the sender during the link establishment phase to request control of the communications link.
repeat	A test that is repeated for a sample to increase confidence in the reported value or to correct an error condition that occurred the first time the test was run.
response	A reply to a query.
results database	Database on the system where the test results and demographic information are stored.
RS-232	A physical layer interface standard for the interconnection of computer systems.
sample type	The kind of sample that you schedule and load on a rack. For the ADVIA Centaur CP system, the sample types are patient, control, and calibrator.

semantics	The relationships of characters or groups of characters to their meanings. This relationship is independent of the way these symbols are interpreted and used.
sender	The computer system that sends data segments during a data link session.
SID	Sample ID. A character string used by the operator of the system to identify a sample.
simplex	CCITT term used to describe a data link duplexity in which both stations may transmit, but only one at a time. Same as ANSI half-duplex.
startup	The events that occur when the ADVIA Centaur CP system initiates the software and the login screen displays. Also called power up or reboot.
stop bits	Bits used for character synchronization by the physical layers of the sending and receiving system. Set to 1 or 2. The default is 1. Two stop bits increase confidence in synchronization, but reduce throughput.
stop-and-wait	A flow control protocol in which the sender transmits a frame of data and then waits for an acknowledgment before transmitting the next frame.
terminate session message	Data link layer message used by the sender during the link release phase to release control of the communications link.
upload	To send data from the system to a remote system.
worklist	A list of scheduled samples, controls, and calibrators, that includes identification and requested tests for each entry.
worklist database	The database on the system in which test requests and demographic information are stored.